

# **The Realized Physical Interior**

Locality, Metric Role, Fields, Particles, Gauge Structure,  
Mass, Vacuum, Time, Causality, and the Standard  
Physics Stack under AASC

A Flagship Synthesis in the AASC Mathematical  
Necessity-Constraint Formalism

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## Abstract

This paper reconstructs the realized physical interior under AASC as the minimal role architecture required for a physical regime to support standing-bearing persistence. AASC is a mathematical necessity-constraint formalism. It is not constructive model-building, not a hidden ontology, not a generative substrate, and not a replacement Lagrangian. Its proof form is eliminative: once a target class is fixed, AASC determines which roles must be present for that target to remain admissibility-bearing, reference-fixed, standing-preserving, and irreversible under lawful continuation. The starting point is therefore not a spacetime manifold, matter inventory, field equation, Hamiltonian, Hilbert space, gauge group, particle spectrum, vacuum sector, or temporal parameter. The starting point is admissibility-governed persistence: a regime in which determinate physical content can stand, be referenced, survive lawful redescription, and undergo irreversible continuation without same-act repair.

The first half of the paper synthesizes and proves the realized physical interior theorem at the role-architecture level. Beginning from the AASC kernel  $K_D = (Adm_D, Stand_D, Ref_D, Irr_D)$ , the physical arc forces non-vacuous realization, bounded standing-localization, local continuation domains, quotient-stable comparison, bounded perturbation transport, fields as persistence carriers, particles as localized persistence classes, gauge structure as boundary-fixed presentation equivalence, mass and inertia as deformation-response roles, vacuum as a minimal non-null no-particle baseline, time as irreversible continuation ordering, and causality as witness-dependent dependence. These roles are not an optional catalogue. They form an interdependent architecture  $I_D$  satisfying coherence, closure, and operational minimality.

The second half develops the standard-physics projections at controlled levels. The AER quotient/constraint architecture supplies coisotropic standing, characteristic redescription, local first-class closure, and multiplier-type Hamiltonian form. Gravity in the AASC sense is the forced constraint role of the admissible physical interior; canonical GR is the paradigmatic downstream realization. A faithful Lorentzian metric projection of that gravity role yields the Einstein field-equation form  $G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{closed}$  as the unique minimal first-response same-scope metric normal form. Quantum predictive structure enters through Hilbert geometry and Schrodinger-form reversible dynamics, with Bell witnesses and measurement records typed as boundary-level quotient structures. Local gauge-invariant Yang-Mills QFT enters as a theorem-object specialization of the field/gauge/vacuum layer. QFT vacuum-energy discipline enters through the equivalence between renormalization-quotient invariance and AASC standing preservation in the homogeneous vacuum sector. The standalone Standard Model structural tuple enters as a downstream structural specialization, while its full empirical parameter table remains a sectoral numerical program.

AASC is mathematical in nature. Its primary proof form is fixed-domain constraint exhaustion and uniqueness up to admissible equivalence. Its mature

predictive force is structural and classificatory, and its downstream numerical program already contains restricted quotient-level and sectoral outputs, including charged-lepton mass-shape extraction, Standard-Model charge-lattice extraction, and a conditional neutrino splitting-ratio readout. These outputs do not amount to full numerical parameter-table closure. The claim of this paper is that the realized physical interior is the minimal admissible role architecture through which, in the role-theoretic sense fixed here, physical theories become about a realized physical regime rather than merely formal presentation.

## Reader's guide and claim-status discipline

The paper is organized in two arcs. Parts I-VI build the realized physical interior. Parts VII-XIII develop the standard-physics projection and specialization architecture. The order matters. Standard physics is not ignored, but neither is it allowed to backflow into the role ladder that licenses it.

Readers should not look here for a constructive substrate out of which physics is generated. The central question is not which primitive object builds fields, particles, metricity, gauge structure, mass, vacuum, time, causality, GR, QFT, or the Standard Model. The central question is which roles cannot be absent once realized physical standing is fixed. AASC answers that question by mathematical necessity constraint: same-scope omission of a required role either fails the target, changes the target, or reintroduces the role under another presentation.

The proof spine is:

$$\text{MetaNecessity} \Rightarrow K_D \Rightarrow \text{AMetric/ATS} \Rightarrow \text{RoleSeq}_D \Rightarrow I_D.$$

The physics spine is:

$$I_D \Rightarrow \{\text{AER}, \text{AASCGravity}, \text{EinsteinProjection}, \text{QuantumInterior}, \text{YMLocalQFT}, \text{ZPERenStanding}, \Sigma_{SM}\}.$$

The arrows in the second display are typed. AER and gravity are quotient/constraint projections; Einstein dynamics is a metric projection; quantum mechanics is a predictive projection; Yang-Mills QFT is a local gauge-field specialization; ZPE is a renormalization-standing equivalence bridge; the Standard Model is a structural specialization. Treating all of these as primitive premises would reverse the dependency order.

## Mathematical status of AASC and the mechanization boundary

This manuscript treats AASC as a mathematical necessity-constraint formalism, not as informal metaphor. Its role vocabulary - admissibility, standing, reference,

irreversibility, fixed scope, redescription, anchor, tensor, and skin - is governed by typed invariance, closure, no-repair, quotient, no-generator, uniqueness, no-backflow, and admissible-equivalence conditions. AASC is mathematical because it fixes a target class, specifies admissible transformations, exhausts same-scope alternatives, and characterizes the surviving structure up to lawful equivalence.

AASC is not constructive in the sense relevant to model-building physics. It does not claim to construct physical objects from lower physical stuff, produce a preferred Lagrangian from primitive matter, or generate fields, particles, metrics, gauge structure, mass, vacuum, time, causality, GR, QFT, or the Standard Model as objects built from a hidden substrate. Its central arguments are necessity arguments. A role survives because every admissible attempt to omit it collapses into nullity, hidden selection, representative privilege, scope change, repair, or skin-only presentation, or else reintroduces the same role under another name.

This proof style therefore differs from analytic existence theory in PDE, constructive differential geometry, symplectic geometry, or algebraic QFT. It is eliminative and rigidity-theoretic rather than generative from a lower analytic substrate. Once a fixed target is declared, every proposed same-scope alternative must either preserve the same invariant structure, collapse to presentation skin, import hidden tensor load, move an anchor, change scope, or violate standing. The realized-physical-interior theorems should therefore be read as mathematical role-exhaustion and structural-rigidity theorems internal to the AASC constraint system.

The formal status of this machinery is not merely promissory. Companion Lean4 formalizations in the AASC corpus cover the kernel and consequence layer through uniqueness, no-generator and non-derivability-from-below consequences, no-repair results, standing conservation, and unique admissible-interior consequences where those artifacts are part of the formal package. This manuscript does not use Lean4 as empirical evidence and does not replace prose proof by mechanization. The mechanized artifacts mark that the constraint layer is formally checkable and that the uniqueness and non-derivability claims are not merely rhetorical factorization choices.

Three proof registers are kept separate. First, the AASC kernel and consequence layer provide the mathematical constraint system. Second, Parts III-VI synthesize and prove the realized physical interior at the role-architecture level by fixed-domain exhaustion. Third, AER, gravity, Einstein projection, quantum structure, Yang-Mills/QFT, zero-point-energy, Standard-Model, charged-lepton, and neutrino results enter as companion theorem families. They are summarized in the main text only to the extent needed for this flagship synthesis; detailed traceability is placed in Appendix A.

## Predictive status of AASC

AASC should not be described as prediction-free. Its primary predictive force is structural and classificatory: it determines which roles can carry standing, which

primitive substitutes fail, which quantities are anchors, tensors, or skin, and which numerical claims require scale, calibration, or exactness certificates. Beyond that structural layer, the downstream numerical program contains restricted quotient-level and sectoral outputs. These do not close the full Standard Model parameter table, but they show that the formalism is not merely verbal or classificatory.

| Predictive layer                       | Current status in this paper                                                                                                  |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Structural prediction                  | Mature: forced role order, realized physical interior, primitive-substitute exclusion, and standard-physics projection typing |
| Classificatory prediction              | Mature: anchor/tensor/skin typing, parameter-status discipline, and zero-point vacuum-energy non-predictivity                 |
| Quotient-level numerical extraction    | Restricted outputs: charged-lepton mass-shape extraction and Standard-Model charge-lattice extraction                         |
| Conditional sectoral numerical readout | Restricted output: neutrino splitting-ratio standing-admission window and one-anchor mass-spectrum display                    |
| Full parameter-table prediction        | Not claimed                                                                                                                   |

The charged-lepton cross-sector arc reports the relation-level mass-shape extraction

$$[1 : 206.7703159727 : 3477.4728371047],$$

and the fixed-scope Standard-Model charge-lattice arc reports the hypercharge-ratio class

$$[1 : 4 : -2 : -3 : -6 : 3]_{\text{norm,sign}}.$$

The neutrino standing-admission arc gives the conditional numerical window

$$0.0299486814926010 \leq \rho_{\nu}^{\text{current}} \leq 0.0299486814926012,$$

with a projective mass-squared shape and one-anchor atmospheric-gap display. The neutrino result is not a zero-anchor derivation of absolute neutrino masses, and it does not derive PMNS numerical coordinates. It is a sectoral numerical readout under declared standing-admission and anchor conditions.

## Claim classes used throughout

**Synthesized and proved at role-architecture level.** The realized physical interior theorem, the coherence/closure/minimality synthesis, the deletion matrix, and the route-exhaustion normal forms are the paper-level synthesis of R1-R12. The component role theorems remain source results; this manuscript integrates them into a single role-architecture proof.

**Companion theorem family.** AER quotient rigidity, AASC gravity, Einstein metric projection, quantum predictive rigidity, local Yang-Mills QFT, ZPE renormalization-standing equivalence, the Standard Model structural tuple, and the restricted numerical-sector outputs are established in companion work and enter here at their declared scope.

**Controlled correspondence.** ADM variables, Lorentzian metrics, Hilbert spaces, local nets, renormalized stress tensors, and Standard Model representations are used as ordinary physical language for downstream structures. They do not function as primitive foundations.

**Downstream parameter target.** Coupling constants, Yukawa eigenvalues, pole masses, CKM/PMNS coordinates, Higgs-sector numerical values, absolute scale anchors, and many cosmological numbers are not silently promoted. They require sector-specific witnesses, calibration ledgers, or exactness certificates.

**Nonclaim.** The paper is not a completed physics encyclopedia. It is a role-architecture and projection theorem for the structural conditions under which later physics becomes physically interpretable.

## Dependency lock

The following table records the single dependency lock used by the main text. Detailed source-to-claim traceability is collected in Appendix A. No later physics theorem object is used to authorize the earlier role architecture that licenses it.

| Layer                   | Function in this manuscript                                                                                                                   |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Meta-necessity          | root reinforcement: irreversible<br>commitment under unrestricted<br>composition requires a<br>standing-preserving admissibility<br>invariant |
| AASC kernel             | construction constraint: admissibility,<br>standing, reference, and irreversibility<br>for governed construction                              |
| AMetric / ATS interface | interface discipline: no primitive<br>metric, coordinate, scale,<br>representative, or skin-as-load route                                     |

| Layer                               | Function in this manuscript                                                                                           |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| RPI role ladder                     | core synthesis: non-nullity, locality, comparison, transport, fields, particles, gauge, mass, vacuum, time, causality |
| AER quotient architecture           | Hamiltonian quotient/constraint specialization                                                                        |
| AASC gravity                        | GR-type constraint role of the admissible physical interior                                                           |
| Einstein metric projection          | minimal first-response Lorentzian metric normal form                                                                  |
| Quantum predictive interior         | Hilbert/Schrodinger predictive structure and boundary record classification                                           |
| Yang-Mills local QFT                | local gauge-invariant pure Yang-Mills QFT theorem object in the declared scope                                        |
| ZPE / renormalization standing      | renormalization-quotient invariance as standing preservation in the homogeneous vacuum sector                         |
| Standard Model structural tuple     | recovered SM structural specialization with parameter table downstream                                                |
| Charged-lepton and neutrino sectors | restricted quotient-level and one-anchor numerical outputs, not full parameter closure                                |

**Constraint-formalism warning.** AASC does not ask which primitive physical substance generates the rest of physics. It asks which roles are mathematically unavoidable once standing-bearing construction and realized physical persistence are fixed. The results below are therefore necessity constraints and same-scope rigidity results, not constructive production rules.

## Part I - Root necessity and admissibility

### 1. The physical interior problem

The paper begins before spacetime, fields, particles, Hilbert spaces, gauge groups, or equations of motion are allowed to function as physical primitives. The target is narrower and more basic: what must be in place for any of those structures to be about a realized physical regime rather than merely a formal presentation?

**Proposition 1.1 - Physical interiority is role-closed standing.** A physical theory that claims determinate same-domain physical content requires a standing-bearing interior architecture:

$$\text{PhysicalTheory}_D \Rightarrow \text{StandingBearingInterior}_D.$$

A mature physical theory usually begins with already functioning objects. A manifold locates events; a field assigns values; a particle label tracks occupants; a Hamiltonian generates evolution; a Hilbert space supports prediction; a gauge group organizes transformations. These structures are indispensable once the relevant roles have standing. They do not, by themselves, explain why the formal structure has physical interiority. A manifold without persistence roles is a container. A field assignment without carrier responsibility is a function. A particle label without localized persistence is a marker. A gauge transformation without boundary-fixed preservation is presentation equivalence only. A time coordinate without retained consequence is an ordering convention.

The realized physical interior is not a new substance added beneath physics. It is the integrated role architecture through which standing-bearing persistence can be non-null, localized, compared, transported, mediated, classified, gauge-stabilized, deformation-responsive, vacuum-supported, temporally ordered, and causally dependent. Primitive spacetime, primitive matter, and primitive formalism may all become excellent representations of this architecture. They cannot be its source unless they already perform the same standing-bearing work.

This target is comparison-class relative. The paper does not classify every possible formal artifact. It classifies regimes that claim determinate same-domain physical content under admissible redescription and irreversible continuation. A system declining that burden may be coherent as another kind of system, but it is not a counterexample to the realized physical interior theorem.

### 2. Meta-necessity of admissibility

Before physical roles are considered, irreversibility itself imposes a constraint. A non-degenerate system that permits unrestricted recombination cannot make irreversible commitments intelligible unless some invariant preserves standing across later states.

**Theorem 2.1 - Meta-necessity of admissibility.**

$$\text{IrrevCommit} \wedge \text{UnrestrictedComp} \wedge \text{NonDeg} \Rightarrow \text{AdmInvariant}.$$

The theorem is not normative. It does not say what a system should preserve. It says that if a decision is genuinely irreversible, then later recomposition cannot be allowed to erase its status without a tracked invariant. If every later state can overwrite an earlier commitment while no invariant records the difference, the earlier decision was reversible in fact. If it cannot be overwritten, then a status-bearing distinction persists. That distinction is already admissibility-relevant.

This result also blocks two tempting foundations. Primitive temporal asymmetry cannot ground admissibility, because an arrow, entropy gradient, psychological direction, or cosmological boundary condition must itself be identified and transported under a standing-preserving scheme. Primitive information cannot ground admissibility either, because entropy, mutual information, algorithmic complexity, Bayesian evidence, and encoding length all presuppose reference stability and redescription-invariant encoding. They are post-admissible measures, not pre-admissible foundations.

The physical arc uses this result only as an upstream lock. It prevents admissibility from being replaced by primitive information, primitive probability, primitive temporal direction, or primitive semantics without reintroducing the same invariant under another name.

### 3. The AASC kernel of non-degenerate construction

The construction-level kernel fixes the formal target inherited by the physical regime. A raw trace of operations is not yet a governed construction. To construct a determinate same-domain object, a system must preserve target identity, evaluate steps, retain standing, and prevent invalid acts from becoming valid retroactively as the same act.

**Theorem 3.1 - Kernel necessity.**

$$\text{TargetAdeq}_D \Rightarrow K_D = (\text{Adm}_D, \text{Stand}_D, \text{Ref}_D, \text{Irr}_D).$$

The four components are mutually supporting. Reference fixes the target of construction. Standing gives a successful act reusable status. Admissibility distinguishes licensed continuation from failure. Irreversibility prevents same-act repair from erasing the fact of failure. Remove any one of them and the target changes: there is either no determinate object, no reusable success, no gate between licensed and unlicensed continuation, or no stable difference between failure and later repair.

This kernel is mathematical in the sense used throughout the paper: the target class is fixed, transformations are typed, and same-domain alternatives are

exhausted by governance-equivalent reintroduction, collapse, or scope change. A physical theory that uses records, observables, lawful redescription, and histories is already operating inside this target class. The realized physical interior is therefore not built beneath the kernel; it is the physical specialization of the kernel's standing-bearing discipline.

#### 4. Constraint formalism and no-backflow

AASC arguments are rigidity arguments over fixed targets. They do not add ontology and do not construct physics from a lower substrate; they remove alternatives that cannot preserve the target without illicit help.

**Proposition 4.1 - Constraint-exhaustion form.** Any proposed primitive substitute for a required role has one of the following outcomes:

$$\text{PrimitiveSubstitute} \Rightarrow \text{Skin} \vee \text{HiddenLoad} \vee \text{ScopeChange} \vee \text{Repair} \vee \text{Collapse}.$$

If the substitute preserves the same standing-bearing work, it is not an alternative but the same role under another presentation. If it changes the target, it is a scope change. If it performs unlicensed work, it imports hidden tensor load or selector authority. If it does no standing-bearing work, it is skin. If it repairs a failed act without declaring repair, it violates irreversibility.

This method controls the later use of standard physics. General relativity, quantum mechanics, Yang-Mills QFT, and the Standard Model are central to the paper, but they enter downstream. A Lorentzian metric may represent a licensed comparison structure; it cannot authorize comparison before the comparison role is fixed. A Hilbert space may represent a predictive quantum interior; it cannot ground record standing before record fixation is fixed. The Standard Model may specialize the field, particle, gauge, mass, and vacuum roles; it cannot be used as a premise for those roles.

The no-backflow condition is:

$$\text{ProjectsTo}(A, X) \wedge \text{Represents}(X, I_D) \not\Rightarrow \text{Generates}(I_D).$$

Downstream success can confirm that a projection is faithful. It cannot retroactively supply the admissibility of the architecture that made the projection meaningful.

## Part II - Interface discipline

### 5. AMetric boundary and no primitive metricity

The AMetric boundary prevents metric, coordinate, temporal, probabilistic, or numerical structure from doing pre-admissible work.

**Theorem 5.1 - Metricity is licensed description.**

$$\text{AMetricBoundary}_D \Rightarrow \neg \text{PrimitiveMetric}_D.$$

A metric expression compares. A coordinate labels. A scale calibrates. A probability measure ranks alternatives. A causal order relates events. Each of these activities presupposes an admitted comparison or ordering structure. At the AMetric boundary, no such structure has standing. If it is used before admission, it functions as a hidden selector. If it is used after admission, it is no longer primitive; it is anchor, tensor, or skin inside an admitted domain.

This distinction also separates AMetric nonmetricity from metric-affine nonmetricity. AMetric nonmetricity is pre-admissible non-selection. Metric-affine nonmetricity is post-metric tensor or connection load inside an already admitted geometric theory. Confusing the two lets a downstream geometric feature masquerade as boundary authority.

The later metric role and Einstein projection are therefore licensed in order: admissibility and standing first, quotient-stable comparison next, faithful metric projection later.

### 6. Anchor, tensor, and skin

Once a domain is admitted, its descriptive elements must be typed by the work they perform.

**Theorem 6.1 - Anchor/tensor/skin trichotomy.** Every same-domain descriptive element is anchor, tensor, or skin:

$$\text{DescElem}_D \Rightarrow \text{Anchor} \vee \text{Tensor} \vee \text{Skin}.$$

An anchor fixes the domain, boundary, comparison class, regularity envelope, or admissibility precondition. Moving an anchor changes the theorem object. A tensor carries standing-bearing invariant content. Moving a tensor changes the physics or role content inside the fixed domain. Skin is presentation surplus: chart, coordinate, notation, representative, gauge, basis, normalization, or display convention. Moving skin changes description but not standing-bearing content.

The trichotomy prevents a common foundational error: giving skin the authority of an anchor or tensor. A coordinate chart cannot create curvature content. A

gauge representative cannot create a new physical state. A Hilbert basis cannot select an outcome. A branch label cannot fix a record. A finite subtraction convention cannot become a cosmological prediction. A MeV display cannot become a zero-anchor mass derivation.

The same trichotomy protects legitimate extensions. Extra scalar fields, torsion, nonlocal operators, stochastic variables, hidden variables, richer QFT sectors, or BSM content are not dismissed. If declared, they are anchors or tensors of a richer scope. If undeclared, they are hidden load. If inert, they are skin.

## 7. Projection and specialization protocol

The standard physics stack enters only after the interior roles have been fixed. Each mature theory object is typed by what it projects or specializes.

**Proposition 7.1 - Projection protocol.** A downstream theory object may represent or specialize the realized physical interior only under no-backflow:

$$\text{ProjectsTo}(A, X) \wedge \text{NoBackflow}(X, I_D).$$

The paper will use several established theorem families. The AER theorem specializes admissibility to Hamiltonian quotient structure on a regular stratum. The gravity theorem identifies the GR-type constraint role of the physical interior. The Einstein projection theorem gives the minimal first-response metric equation inside a faithful Lorentzian projection. The quantum theorem fixes Hilbert/Schroedinger predictive structure and record quotients. The Yang-Mills theorem supplies a local gauge-invariant QFT realization. The zero-point-energy results connect renormalization quotienting to standing preservation. The Standard Model theorem supplies the structural tuple.

Each object has a scope guard. AER is local on a regular stratum. Einstein dynamics is first-response metric projection, not all gravitational dynamics. Quantum predictive rigidity is not category equivalence with GR. Yang-Mills QFT is a local pure-gauge theorem object, not all QFT. The zero-point-energy result is homogeneous vacuum-sector quotient discipline, not a solution to naturalness. The Standard Model result is structural, not full parameter-table closure.

## Part III - Realized persistence, locality, and transport

The following sections give the realized-physics ladder. They are written as role-exhaustion results inside the AASC mathematical constraint system. The individual role theorems are sourced in the R1–R12 arc and are consolidated here as a single physical-interior argument; the source-theorem index in Appendix A records exact lineage. The point of this part is not to rename familiar physical objects. It is to isolate the standing-bearing work each familiar object must perform before it can be used as part of a realized physical regime.

### 8. Non-vacuous realized persistence

**Theorem 8.1 - Non-null realization.** If a domain is to count as a realized physical persistence regime, then its persistence cannot be reconstruction-null:

$$\text{RealPers}_D \Rightarrow \text{NonVacRec}_D.$$

**Proof.** A physically realized persistence regime must leave some reconstruction-bearing content available to later admissible use. If all alleged persistence can be removed without changing any standing-bearing reconstruction, then nothing has persisted in the relevant sense: there is only an empty label, a vacuous admission, or a formal mark. Conversely, if later operations can recover, compare, damage, restore, or continue something as the same physical content, then a non-null reconstruction-bearing residue is already present. The null alternative therefore does not give a weaker physical realization; it removes the target.

This first role is intentionally austere. It does not yet supply spacetime, fields, particles, metric comparison, or dynamics. It supplies only the floor on which those later roles can have work to do. A primitive declaration that “there is a physical system” is insufficient unless the declaration carries non-vacuous reconstruction-bearing content.

**Deletion consequence.** Without  $\text{NonVacRec}_D$ , every later role is forced to act on an empty target. Locality becomes a place for nothing, fields become assignments without persistence content, and vacuum collapses into nullity rather than a no-particle baseline.

**Physical reading.** Non-null realization is the minimum difference between a formal description and a physically realized regime. It is the floor of  $I_D$ , not a matter ontology.

### 9. Bounded standing-localization and physical locality

**Theorem 9.1 - Origin of physical locality.** If realized persistence is subject to restoration, degradation, or failure assessment, then the regime admits a bounded standing-localization class:

$$\text{RealPers}_D \wedge \text{RestCon}_D \Rightarrow [B_D]_{\equiv B}.$$

**Proof.** Restoration and failure cannot be evaluated against an unlocalized totality. A persistence burden must have some bounded locus at which damage, recovery, retained identity, and loss are assessed. If the burden is everywhere at once, then every perturbation is global and there is no local restoration problem. If it is nowhere, then there is no persistence burden to preserve. If a representative point is selected as primitive, the proof has imported a locator rather than derived locality. The admissible residue is therefore not a privileged coordinate point, but a bounded equivalence class of standing-bearing burden.

This result makes locality prior to distance. Distance presupposes a comparison carrier; bounded standing-localization requires only that persistence burden not dissolve into globality or nullity. A metric may later represent such local structure, but it does not create it.

**Deletion consequence.** Remove  $[B_D]_{\equiv B}$ , and the regime has no bounded locus for burden, restoration, failure, or retained identity. Later field and particle roles must then import a spatial support by hand.

**Physical reading.** Physical locality begins as bounded standing-burden. Manifold points, regions, supports, and local algebras are later presentations of that more primitive role.

## 10. Continuation loci and local domains

**Theorem 10.1 - Local-domain organization.** Bounded loci become usable physical structure only when they are organized by admissible continuation:

$$[B_D]_{\equiv B} \wedge \text{BoundaryFix}_D \Rightarrow (\text{Adj}_D, N_D, \text{LCD}_D).$$

**Proof.** A single bounded locus is not yet a local domain. Physical continuation requires knowing which loci can be lawfully continued with which others, which boundary traces are compatible, and which neighborhoods preserve standing under local change. Adjacency  $\text{Adj}_D$  records the minimal continuation-relevant contact relation. Neighborhoods  $N_D$  record the bounded families available for local comparison and continuation. Local continuation domains  $\text{LCD}_D$  collect the loci over which standing-preserving operations can be evaluated without changing scope.

A primitive topology would be too strong at this stage. It would impose open sets and continuity before the admissibility role of locality has been fixed. The AASC route instead derives the local-domain role from boundary-compatible continuation of bounded standing loci.

**Deletion consequence.** Without  $(\text{Adj}_D, N_D, \text{LCD}_D)$ , a bounded burden has no lawful local environment. Transport has no path discipline, fields have no

local carrier domain, and causality has no local witness structure.

**Physical reading.** Local domains are the pre-metric architecture that later supports neighborhoods, regions, local algebras, and local field assignments.

## 11. Metric role from continuation comparison

**Theorem 11.1 - Quotient-stable comparison.** Once local continuation domains are available, metric role begins as stable comparison across those domains:

$$(Adj_D, N_D, LCD_D) \Rightarrow \mathbf{Cmp}_D.$$

**Proof.** Local domains can be adjacent and continuable without yet being comparable. To evaluate deformation, transport reach, relative burden, or response, the regime needs comparisons that survive lawful redescription. A bare coordinate difference does not suffice: it can change under presentation. A graph distance does not suffice unless its standing under continuation is already fixed. A numerical metric tensor is also too late, because it presupposes a licensed metric projection.

The admissible object is the quotient-stable comparison carrier  $\mathbf{Cmp}_D$ . It records comparative content that descends through redescription and can be reused by later transport, mass, and causal roles. This is why the manuscript separates  $\mathbf{Cmp}_D$  from  $\mathbf{Cau}_D$ : comparison and causal dependence do different work.

**Deletion consequence.** Without  $\mathbf{Cmp}_D$ , the regime may have local domains but no stable way to compare deformation, reach, transport, or response. Metric language would then enter as primitive notation rather than licensed physical role.

**Physical reading.** Metricity first appears as comparison standing, not as a pre-given distance function. Lorentzian metrics, Riemannian metrics, and other numerical comparison structures are later metric projections.

## 12. Bounded perturbation transport

**Theorem 12.1 - Bounded transport.** A persistence regime with local domains and comparison must constrain how perturbation burden moves:

$$(LCD_D, \mathbf{Cmp}_D) \wedge \mathbf{Perturb}_D \Rightarrow \mathbf{BoundTrans}_D.$$

**Proof.** A perturbation that can affect any domain without a bounded transport relation destroys the local role already obtained. It makes standing failure pathless and restoration nonlocal. A perturbation that cannot be transported at all is equally unusable: no local response, interaction, or mediated effect can be represented. The admissible middle is bounded perturbation transport: transport

whose reach is organized by the local-domain and comparison structures already fixed.

This is not yet a numerical light cone, a propagator, or a finite speed. Those are later physical presentations. At this stage the result is structural: perturbation burden must be carried through admissible local-continuation routes rather than appearing globally, pathlessly, or by hidden selector.

**Deletion consequence.** Without  $\text{BoundTrans}_D$ , later fields have no disciplined way to mediate persistence, causal dependence becomes pathless, and interactions become either global bookkeeping or scope-changing import.

**Physical reading.** Bounded transport is the structural precursor of finite-propagation constraints, support properties, propagators, and causal cones.

## Part IV - Fields, particles, gauge, mass, and vacuum

### 13. Fields as persistence carriers

**Theorem 13.1 - Field-carrier mediation.** Local persistence under bounded transport requires distributed carrier mediation:

$$(LCD_D, \text{Cmp}_D, \text{BoundTrans}_D) \Rightarrow \Phi_D.$$

**Proof.** Local domains and bounded transport specify where and how persistence burden can move, but they do not yet say what carries that burden. A primitive field value is only an assignment unless it is responsible for preserving standing-bearing content across local continuation. A global substrate overdoes the work by erasing locality; isolated point values underdo it by failing to mediate transport.

The field role  $\Phi_D$  is the distributed persistence carrier: it assigns carrier profiles over local continuation domains in a way compatible with comparison and bounded transport. A field is therefore not admitted because physics usually uses fields. It is admitted because persistence across local domains needs a carrier that is neither a hidden object core nor a mere record of outcomes.

**Deletion consequence.** Without  $\Phi_D$ , perturbations may have routes but no domain-internal mediator. Particle identity becomes a label, gauge equivalence has no content to preserve, and vacuum becomes an empty container or nullity.

**Physical reading.** Classical fields, quantum fields, curvature fields, and local operator assignments are mature representations of the carrier role, not primitive sources of it.

### 14. Particles as localized persistence classes

**Theorem 14.1 - Localized persistence identity.** Once field carriers are available, localized objecthood is forced as a persistence class rather than a primitive point or name:

$$(\Phi_D, \text{BoundTrans}_D, \text{Cmp}_D, LCD_D) \Rightarrow \Pi_D.$$

**Proof.** A field carrier can mediate persistence without yet isolating stable localized identity. To speak of a particle, the regime needs a repeatable class of localized carrier configurations that survives admissible redescription, comparison, and transport. A point occupant is too little: it gives location without persistence class. A hidden object core is too much: it imports identity without showing how the carrier regime preserves it. A measurement label records an event but does not by itself carry objecthood.

The particle role  $\Pi_D$  is therefore the equivalence class of localized persistence configurations. It is carried by field mediation and stabilized by boundary and transport conditions. Objecthood is earned by persistence-class behavior, not assumed by naming.

**Deletion consequence.** Without  $\Pi_D$ , the interior has fields and transport but no localized identity bearer. Mass has no target, measurement has no stable occupant, and SM representation slots become labels without particle-role content.

**Physical reading.** Particles in QFT, excitations, localized states, and representation-theoretic species are later specializations of  $\Pi_D$ .

## 15. Gauge as boundary-fixed presentation equivalence

**Theorem 15.1 - Gauge role.** Field-carrier and particle-role content admit presentation multiplicity, and that multiplicity must be disciplined by boundary-fixed equivalence:

$$(\Phi_D, \Pi_D, \text{BoundTrans}_D, \text{Cmp}_D, \text{BoundaryFix}_D) \Rightarrow G_D.$$

**Proof.** Once the regime contains fields and localized persistence classes, the same physical content can appear under different local presentations. Treating every presentation as a distinct physical state creates false multiplicity. Treating all presentation differences as arbitrary convention erases real transition and boundary constraints. The gauge role is the admissible middle: boundary-fixed presentation equivalence preserving carrier and particle content.

This result rejects an ontic reading of gauge multiplicity without demoting gauge to a disposable notation. Gauge transformations are lawful redescrptions; gauge connections, holonomies, and charges can later carry tensor content when they affect standing-bearing transport or boundary comparison. The role  $G_D$  fixes the equivalence discipline before any specific group or coupling is introduced.

**Deletion consequence.** Without  $G_D$ , field and particle presentations either proliferate into false physical multiplicity or collapse into ungoverned convention. Charges, gauge transport, and SM gauge structure then have to be imported from outside the fixed domain.

**Physical reading.** Gauge groups, bundles, connections, curvature, and local gauge-invariant algebras are specializations of boundary-fixed presentation equivalence.

## 16. Mass and inertia as deformation-response roles

**Theorem 16.1 - Deformation response and inertia.** Localized persistence classes inside a field-carrier regime require a response profile under admissible deformation and transport-state change:

$$(\Phi_D, \Pi_D, G_D, \text{BoundTrans}_D, \text{Cmp}_D) \Rightarrow M_D = (\mu_D, \iota_D).$$

**Proof.** A particle persistence class can remain identifiable only if the regime has a way to register how that class responds to standing-threatening deformation. A primitive scalar mass does not supply this work. It states a number but does not explain why the localized class resists deformation, how response is compared, or how transport-state change is constrained. Conversely, a purely relational label cannot support inertial response.

The mass role  $\mu_D$  is the deformation-response profile of a persistence class. The inertia role  $\iota_D$  is the transport-facing restriction of that profile. Both require comparison, carrier mediation, particle identity, and gauge-compatible presentation discipline. This is why the later stress-energy ledger can be typed as metric-facing extraction from source roles rather than primitive mass-energy substance.

**Deletion consequence.** Without  $M_D$ , the interior has localized classes but no standing-bearing profile for deformation, compression, inertial response, or metric-facing source extraction. Mass values would then enter as untyped scalars.

**Physical reading.** Inertial parameters, mass terms, Higgs/Yukawa mass structures, and stress-energy contributions are downstream presentations of deformation-response standing.

## 17. Vacuum as minimal concrete persistence baseline

**Theorem 17.1 - Vacuum baseline.** A field/particle/gauge/mass regime needs a non-null no-particle baseline:

$$(\Phi_D, \Pi_D, G_D, M_D, \text{BoundTrans}_D) \Rightarrow \Omega_D.$$

**Proof.** Removing localized particle-class excitations does not remove the carrier regime, boundary trace, transport possibilities, comparison structure, gauge compatibility, or deformation-response baseline. A no-particle condition therefore cannot be identified with nullity. Nor may it be treated as a primitive substrate, since that would supply standing without the carrier and quotient roles already derived.

The vacuum role  $\Omega_D$  is the minimal concrete persistence baseline that remains when particle excitations are absent. It supports the possibility of field-carrier structure, local response, gauge standing, and later quantum vacuum sectors. It also explains why absolute zero-point scalar offsets do not automatically carry source standing: a scalar vacuum number is not the vacuum role.

**Deletion consequence.** Without  $\Omega_D$ , no-particle stages collapse either to nullity or to substrate import. QFT vacuum sectors, homogeneous metric ledgers, and baseline response quantities lose their role anchor.

**Physical reading.** Vacuum vectors, vacuum sectors, ground states, Higgs/vacuum structure, and cosmological homogeneous ledgers are mature specializations of a non-null no-particle persistence baseline.

## Part V - Time, causality, records

### 18. Time as irreversible continuation ordering

**Theorem 18.1 - Temporal ordering.** A realized interior with vacuum, fields, particles, gauge, mass, transport, and comparison supports physical time only as ordered retained consequence:

$$(\Omega_D, \Phi_D, \Pi_D, G_D, M_D, \text{BoundTrans}_D, \text{Cmp}_D) \Rightarrow T_D.$$

**Proof.** The kernel already contains irreversibility of commitment, but physical time requires more than kernel finality. A physical interior must order standing-bearing stages so that retained consequence, failure, restoration, and continuation can be located in a history. A primitive time parameter would merely label states unless it preserved the standing-bearing content of those stages. Observer chronology is likewise insufficient, because it presupposes record fixation rather than grounding it.

The temporal role  $T_D$  is the ordering of admissible continuation stages by irreversible retained consequence. It may later be represented by coordinate time, proper time, Hamiltonian evolution, thermodynamic arrows, or quantum unitary parameters, but none of those representations is primitive at the role-ladder stage.

**Deletion consequence.** Without  $T_D$ , the interior has no ordered lineage of standing-preserving change. Records cannot be fixed as earlier or later, failure cannot be retained, and causal dependence has no temporal carrier.

**Physical reading.** Time in standard physics is a representation of ordered continuation; the role is prior to any specific clock, foliation, Hamiltonian parameter, or entropy gradient.

### 19. Causality as witness-dependent dependence

**Theorem 19.1 - Causal dependence.** Temporal order becomes causality only when dependence is witnessed across that order:

$$T_D \wedge \text{WitnessDep}_D \Rightarrow \text{Cau}_D.$$

**Proof.** Not every earlier stage is a cause of every later stage. Temporal order supplies before-and-after; causality requires a dependence relation that survives admissible witness comparison. A primitive causal arrow would either duplicate time, import hidden influence, or assert dependence without a witness. A purely metric light cone is also too late, since metric causal structure is a later projection of comparison, transport, and temporal roles.

The causal role  $\text{Cau}_D$  identifies those ordered pairs whose later standing-bearing content depends on earlier content through an admissible witness. Thus

$$e_i \rightsquigarrow_D e_j \Rightarrow e_i \prec_D e_j,$$

but order alone does not imply dependence. This distinction allows the later GR, QFT, and measurement-record sections to treat causal cones, propagators, Bell witnesses, and records without confusing correlation, order, and cause.

**Deletion consequence.** Without  $\text{Cau}_D$ , the interior has chronology and transport but no typed dependence. Effects become pathless influence, mere succession, or externally imposed causal bookkeeping.

**Physical reading.** Relativistic causal structure, propagator support, causal dependence in field theory, and evidential dependence in measurement are all specializations of witness-dependent ordered dependence.

## 20. Measurement records and empirical standing

**Theorem 20.1 - Record-fixation quotient.** Empirical records become standing-bearing only through admissible record fixation:

$$\text{RecordUse}_D \Rightarrow A_{\text{Ref}} / \simeq_{MR}.$$

**Proof.** A measurement mark is not yet evidence. To function as evidence, it must fix what it records, survive lawful redescription, be reusable in later reasoning, and retain the difference between admissible and failed record use. If record assignments can be freely permuted while preserving all standing-bearing profiles, then the distinct assignments are skin and must be quotiented. If two assignments differ in standing-bearing profile, the difference is tensor content and cannot be dismissed as convention.

The measurement-record quotient  $A_{\text{Ref}} / \simeq_{MR}$  is the standing-bearing content space for admissible record use. It also controls empirical constants. A number displayed by a theory may be a calibration anchor, a tensor readout, a skin coordinate, a conditional one-anchor display, or a deferred parameter target. It is not automatically a prediction merely because a formalism can print it.

**Deletion consequence.** Without record-fixation quotienting, empirical comparison becomes assignment-relative. Fit can masquerade as derivation, calibration can masquerade as zero-anchor prediction, and measurement outcomes can be reselected after the fact.

**Physical reading.** This record layer supports the later quantum measurement analysis, the zero-point-energy witness ledger, and the Standard Model parameter-status discipline.

## Part VI - Realized physical interior synthesis

### 21. Definition of the realized physical interior

The preceding role ladder now closes into one object. The realized physical interior is not a spatial inside, a manifold interior, a set-theoretic interior, or a primitive container. It is the standing-bearing internal role organization of the regime.

**Definition 21.1 - Realized physical interior.** For a fixed admissible realized persistence regime  $D$ , define

$$I_D = (\text{NonVacRec}_D, [B_D]_{\equiv B}, \text{Adj}_D, N_D, LCD_D, \text{Cmp}_D, \text{BoundTrans}_D, \Phi_D, \Pi_D, G_D, M_D, \Omega_D, T_D, \text{Cau}_D).$$

The order records dependency rather than temporal sequence. The floor is non-null realization and vacuum baseline. The local layer is bounded standing-localization, adjacency, neighborhoods, and local continuation domains. The comparison and transport layer is  $\text{Cmp}_D$  together with  $\text{BoundTrans}_D$ . The carrier/object layer is  $\Phi_D$  and  $\Pi_D$ . The presentation/response layer is  $G_D$  and  $M_D$ . The ordering layer is  $T_D$  and  $\text{Cau}_D$ .

The definition is deliberately role-first. Later formalisms may represent these roles by manifolds, metrics, local algebras, fields, Hilbert spaces, bundles, stress-energy tensors, gauge groups, or Lagrangians. Those formalisms are not denied. They become physically interpretable by occupying the role architecture rather than replacing it.

### 22. The interior is not a list

**Theorem 22.1 - Role interdependence.** The components of  $I_D$  are mutually supporting rather than independent additions:

$$\text{RoleSeq}_D \Rightarrow \text{Interdependent}(I_D).$$

Fields require local continuation domains and bounded transport; otherwise carrier mediation has no local support or transport discipline. Particles require fields; otherwise localized identity collapses into point labels, path labels, records, or hidden object cores. Gauge requires field and particle role content; otherwise presentation equivalence preserves nothing physical. Mass and inertia require particle identity, field mediation, comparison, bounded transport, and gauge compatibility; otherwise deformation response becomes scalar import. Vacuum requires field, gauge, transport, comparison, and deformation baseline roles; otherwise no-particle structure becomes nullity or substrate. Time and causality require record-fixable standing stages, bounded transport, and non-null baseline support.

The interdependence theorem is the conceptual heart of the paper. A reader may accept every role locally and still think the capstone is a catalogue. It is not. Later roles use earlier roles, and earlier roles acquire physical maturity through later closure. Locality becomes physically meaningful when it can host fields, particles, gauge, mass, vacuum, time, and causality. Vacuum becomes physically meaningful because it is the baseline of the same carrier and transport architecture. Time becomes physically meaningful because it orders retained standing-bearing consequences within the non-null regime.

This also blocks primitive starts. A field-first ontology must already supply local domains, transport, particle identity, gauge discipline, mass response, vacuum baseline, and ordering. A particle-first ontology must already supply carrier mediation. A metric-first ontology must already supply admissible comparison and standing. Each start can become a specialization of  $I_D$ ; none is the source of the whole interior.

## 23. Coherence theorem

**Theorem 23.1 - Interior coherence.** If the role sequence is jointly instantiated, the roles preserve one standing-bearing physical content under admissible continuation:

$$\text{RoleSeq}_D \Rightarrow \text{Coher}_D(I_D).$$

Coherence is not mere syntactic consistency. It means that the bounded locus, local domain, comparison carrier, transport burden, field carrier, particle class, gauge equivalence, deformation response, vacuum baseline, temporal order, and causal dependence all concern the same physical content. If a later role silently switches target, the interior fractures.

The proof follows the dependency chain. Non-vacuous realization fixes that there is a physical target to preserve. Localization fixes where standing burden is evaluated. Local domains fix the neighborhoods over which continuation can be assessed. Comparison fixes quotient-stable relational content. Bounded transport fixes how perturbations move without becoming global or pathless. Fields provide distributed carriers over those domains. Particles provide localized identity classes in those carriers. Gauge identifies presentation variants preserving the same role content. Mass records deformation-response of those identity classes. Vacuum supplies the no-particle baseline for the same carrier/transport/gauge architecture. Time orders retained consequences. Causality identifies witness-dependent dependence inside the order.

If any layer targets a different object, the disagreement is not a coherent interior. It is representative privilege, identity reset, cross-domain transfer, hidden selector import, or scope change.

## 24. Closure theorem

**Theorem 24.1 - Interior closure.** The role package  $I_D$  is role-closed for realized physical persistence:

$$\text{RoleSeq}_D \Rightarrow \text{Closure}_D(I_D).$$

Closure has a positive and a negative side. Positively, the interior supplies non-null support, bounded localization, local-domain organization, comparison, bounded transport, carrier mediation, localized identity, gauge-compatible presentation discipline, deformation response, vacuum baseline, temporal order, and causal dependence. Negatively, it excludes null collapse, hidden selector import, cross-domain transfer, skin-only representation, identity reset, gauge artifact, pathless transport, primitive time import, primitive metric import, and vacuum/substrate confusion.

Closure does not mean the end of physics. It does not supply every law, parameter, boundary condition, or empirical model. It means no further primitive source is needed to make the physical interior interpretable at the role level. Later theories specialize, project, enrich, or parameterize the architecture; they do not replace the role conditions that make them physical.

The closure theorem also establishes a classification standard. A proposed addition must declare whether it is anchor, tensor, skin, richer scope, effective scope, boundary data, or parameter target. If it changes the theorem object, it is not a same-scope rival. If it changes nothing, it is skin. If it changes something without being declared, it is hidden load.

## 25. Operational minimality theorem

**Theorem 25.1 - Operational minimality.** No major role family of  $I_D$  can be deleted while preserving realized physical interiority:

$$\text{RoleSeq}_D \Rightarrow \text{Minimal}_D(I_D).$$

Minimality is evaluated by deletion. Remove non-null realization and the interior has no physical support. Remove bounded localization and local burden, failure, and restoration become global or diffuse. Remove local domains and transport has no domain structure. Remove comparison and metric/deformation roles become primitive uses. Remove bounded transport and perturbation becomes pathless. Remove fields and persistence has no distributed carrier. Remove particles and localized identity disappears. Remove gauge and presentation variation becomes artifact or false multiplicity. Remove mass and deformation response becomes scalar bookkeeping. Remove vacuum and no-particle stages collapse into nullity or substrate. Remove time and no irreversible continuation order remains. Remove causality and dependence is externalized.

The theorem does not assert that no refinements, equations, constants, state spaces, or sectoral structures may be added. It asserts that the major role families cannot be removed while preserving the same target. A counterexample must preserve standing-bearing realized physical persistence while deleting one of these roles and avoiding the listed failure.

## 26. Main realized physical interior theorem

**Theorem 26.1 - Realized physical interior synthesis.** Let  $D$  be a fixed admissible realized persistence regime. If the AASC kernel is instantiated and the role sequence is jointly realized, then  $D$  induces the realized physical interior satisfying coherence, closure, and operational minimality:

$$K_D \wedge \text{FixedAdmRealPers}_D \wedge \text{RoleSeq}_D \Rightarrow \text{RPI}_D \wedge \text{Coher}_D(I_D) \wedge \text{Closure}_D(I_D) \wedge \text{Minimal}_D(I_D).$$

The role sequence is

$$\text{NonVacRec}_D \wedge [B_D]_{\equiv B} \wedge (\text{Adj}_D, N_D, \text{LCD}_D) \wedge \text{Cmp}_D \wedge \text{BoundTrans}_D \wedge \Phi_D \wedge \Pi_D \wedge G_D \wedge M_D \wedge \Omega_D \wedge (T_D, \text{Cau}_D).$$

The theorem's substantive content is not tuple formation. It is that the derived roles preserve a common standing-bearing physical content, close the role operations needed by realized physical persistence, and survive deletion testing as a minimal family. The first half of the paper therefore synthesizes and proves the realized physical interior theorem at the role-architecture level. The second half brings in mature physics theorem objects as projections or specializations of this architecture, not as primitive sources for it.

## Part VII - AER quotient and constraint architecture

The realized physical interior theorem identifies the role architecture required for standing-bearing physical persistence. Part VII explains how that role architecture is represented when the physical target is specialized to regular-stratum Hamiltonian observable evaluation. This is the first mature physics specialization in the paper. It shows how the role ladder is represented when physical content is evaluated through a quotient of admissible presentations, physical observables, and lawful continuation.

The AER quotient/constraint specialization has the following logical order:

$$K_D \Rightarrow \text{AASCIInterface}_D \Rightarrow \text{AERInterface} \Rightarrow \text{CoisotropicStanding} \Rightarrow \text{FirstClassClosure} \Rightarrow \text{MultiplierForm}.$$

The following sections give the local form of that specialization. The details are not primitive assumptions of the realized physical interior; they are the regular Hamiltonian realization of the interior once the AER comparison class is selected.

The standard mathematical background for this part is the constrained-Hamiltonian tradition associated with Dirac, Bergmann, Marsden-Weinstein reduction, and the later systematic treatment of gauge constraints. The paper does not use that tradition as a premise for the AASC role ladder. It uses it as the ordinary mathematical setting in which the AER quotient theorem object can be read by a physics audience: standing surfaces correspond to constraint surfaces, admissible redescription corresponds to characteristic/gauge motion, and physical observables are quotient-level quantities.

**Source note.** The AER companion theorem family is indexed in Appendix A. The main text states the local interface and proof chain rather than displaying the source ledger in-line.

### 27. Why physical interior requires quotient architecture

A physical interior cannot identify its content with a single presentation. Coordinates, gauges, bases, regularizations, representative extensions, and local labels may vary while preserving the same physical content. The interior therefore needs a quotient architecture: a way to distinguish standing-bearing tensor content from presentation skin while preserving lawful continuation.

**Definition 27.1 (physical quotient presentation).** A physical quotient presentation is a tuple

$$Q = (C, \pi : C \rightarrow S_{phys}, \mathcal{O}_{phys}, \sim, \text{Cont}),$$

where  $C$  is the standing surface of admissible states,  $\pi$  is the quotient map to the physical state space,  $\mathcal{O}_{phys}$  is the algebra of quantities constant on admissible-redescription classes,  $\sim$  is the equivalence relation generated by lawful redescription, and  $\text{Cont}$  is the continuation rule whose observable effects are evaluated on the quotient.

The quotient is not an optional interpretive gloss. If two presentations are related by admissible redescription and the theory nevertheless treats them as distinct physical contents, it has introduced representative privilege. If it fails to distinguish presentations whose observable and continuation behavior differ, it has collapsed tensor content into skin. Thus quotient identity is the physical expression of reference and standing under redescription.

**Proposition 27.2 (quotient necessity for the physical interior).** If a realized physical interior supports lawful redescription and reusable same-domain physical content, then standing-bearing content factors through a quotient by admissible redescription.

**Proof.** Let  $x, y \in C$  be presentations related by a lawful redescription. If  $x$  and  $y$  are assigned different standing-bearing physical content while all admitted witnesses and continuations agree, then the difference is carried only by the presentation. That is skin promoted to tensor authority. Conversely, if  $x$  and  $y$  differ in some admitted observable, redescription, or continuation effect, then they are not the same physical content and cannot be quotient-identified without destroying reference. The quotient is therefore exactly the carrier of standing-bearing identity under lawful presentation change.

The AER theorem object is the Hamiltonian version of this requirement. It specializes quotient identity to observable Hamiltonian evaluation on a regular stratum. This is why it is the natural bridge from  $I_D$  to GR-type constraint structure: GR's constraint and gauge features are not defects of formulation but the Hamiltonian expression of quotient-bearing physical content.

## 28. AER interface

The AER interface starts with ordinary symplectic data but does not assume constraint functions, first-class closure, or a reduced symplectic form. Let  $(S, \omega)$  be a finite-dimensional symplectic manifold on a regular stratum. The regularity restriction means that the standing surface is embedded of constant codimension, the quotient map is a smooth submersion, and the local redescription pseudogroup acts smoothly where used.

**Definition 28.1 (standing surface and quotient).** The standing surface is a smooth embedded submanifold

$$C \subset S.$$

Standing is binary:  $\text{Stand}(x) = 1$  iff  $x \in C$ . A local pseudogroup of admissible

redescriptions acts on  $C$  and generates an equivalence relation. The smooth local quotient is

$$\pi : C \rightarrow S_{phys} = C / \sim .$$

**Definition 28.2 (redescription distribution).** The infinitesimal redescription distribution is

$$D_x = \ker(d\pi_x) \subset T_x C.$$

Because  $\pi$  is a smooth submersion,  $D$  is the vertical bundle of the quotient. Hence  $D$  is involutive: if  $X$  and  $Y$  are vertical vector fields, then  $d\pi[X, Y] = [0, 0] = 0$ , so  $[X, Y]$  is vertical.

**Definition 28.3 (physical observables).** A physical observable is a smooth function  $O : C \rightarrow \mathbb{R}$  that is constant along admissible-redescription classes. Equivalently,

$$O = \tilde{O} \circ \pi$$

for some smooth  $\tilde{O} : S_{phys} \rightarrow \mathbb{R}$ .

The AER interface adds three observable-completeness clauses. These are not arbitrary modeling postulates; they are the local Hamiltonian form of admissibility, reference, and standing for the target theorem object.

**AER-Obsmin.** If  $O$  and  $P$  are physical observables on  $C$ , then the Poisson bracket of arbitrary ambient extensions restricts to  $C$  in a way independent of the chosen extensions and descends to a bracket on  $S_{phys}$ . In particular, an extension of the zero physical observable brackets to zero with every physical observable after restriction to  $C$ .

**AER-Sep.** Physical observables separate quotient-transverse directions. More precisely, if two quotient classes are distinct, some physical observable distinguishes them; and at each  $x \in C$ , the differentials of physical observables span

$$\text{Ann}(D_x) \subset T_x^* C.$$

**AER-Max.** No nonzero presymplectic class remains invisible to all physical observables. Equivalently, if  $v \in T_x C$  and  $v \notin \ker(\omega_{C,x})$ , then some physical observable  $O$  satisfies  $dO_x(v) \neq 0$ , where  $\omega_C = \iota^* \omega$  is the pullback of the ambient symplectic form to  $C$ .

AER-Obsmin supplies quotient-level Poisson closure. AER-Sep supplies quotient detectability. AER-Max supplies observable maximality. If AER-Max is omitted, one can have a quotient whose physical observables are closed and separated

relative to the chosen quotient but which still leaves a hidden presymplectic direction invisible to those observables. The source AER paper includes an explicit local witness of this failure using  $S = \mathbb{R}^4$  with  $C = \{p_1 = 0\}$  and a quotient that forgets a non-characteristic direction. The paper does not need the witness as a premise, but it needs the lesson: AER-Max is the minimal observable-completeness clause preventing hidden same-scope Hamiltonian remainder.

**Theorem 28.4 (AER interface theorem).** For regular-stratum observable Hamiltonian evaluation, the tuple

$$(C, \pi, D, \omega_C, \mathcal{O}_{phys})$$

together with AER-Obsmin, AER-Sep, and AER-Max is the local observable quotient interface required for same-scope admissibility-bearing Hamiltonian content.

**Argument.** Standing supplies  $C$ . Reference and lawful redescription supply  $\pi$  and  $D$ . Internal closure of outcome-relevant content supplies physical observables. Hamiltonian evaluation requires the observable bracket to be defined on quotient content rather than on arbitrary ambient extensions, giving AER-Obsmin. Determinate physical reference between quotient classes requires observable separation, giving AER-Sep. Finally, the fixed-domain no-hidden-remainder discipline excludes presymplectic directions that would carry Hamiltonian content while remaining invisible to all physical observables, giving AER-Max. Thus the interface is the Hamiltonian specialization of the admissibility package, not a disguised assumption of GR.

**Source note.** Exact AER theorem labels for the interface, coisotropic forcing, first-class closure, multiplier form, and fixed-domain closure are collected in Appendix A.

## 29. Coisotropic forcing and characteristic redescription

The key local result is the identification of admissible redescription with the characteristic distribution of the pulled-back symplectic form.

Write

$$\ker(\omega_{C,x}) = \{v \in T_x C : \omega_{C,x}(v, w) = 0 \text{ for all } w \in T_x C\}.$$

Also write the ambient symplectic orthogonal of  $T_x C$  as

$$(T_x C)^{\perp_\omega} = \{\xi \in T_x S : \omega_x(\xi, w) = 0 \text{ for all } w \in T_x C\}.$$

**Lemma 29.1 (characteristic capture).** Under AER-Obsmin and AER-Sep,

$$(T_x C)^{\perp_\omega} \subseteq D_x.$$

Consequently  $C$  is coisotropic.

**Argument.** Let  $\xi \in (T_x C)^{\perp_\omega}$ . Since  $\omega_x(\xi, \cdot)$  vanishes on  $T_x C$ , choose a local smooth function  $H$  with  $H|_C = 0$  and  $dH_x = \omega_x(\xi, \cdot)$ . Its Hamiltonian vector at  $x$  is  $X_H(x) = \xi$ . Let  $O$  be any physical observable and  $\tilde{O}$  any ambient extension. Since  $H|_C$  is the zero physical observable, AER-Obsmin gives  $\{H, \tilde{O}\}|_C = 0$ , so  $d\tilde{O}_x(\xi) = 0$ . Applying the same zero-observable neutrality to functions vanishing on  $C$  first shows  $\xi \in T_x C$ . Then  $dO_x(\xi) = 0$  for all physical  $O$ . By AER-Sep, the differentials of physical observables span  $\text{Ann}(D_x)$ , so  $\xi$  annihilates  $\text{Ann}(D_x)$  and hence  $\xi \in D_x$ . Since  $D_x \subset T_x C$ , the inclusion  $(T_x C)^{\perp_\omega} \subset T_x C$  proves coisotropy.

**Lemma 29.2 (no hidden presymplectic remainder).** Under AER-Max,

$$D_x \subseteq \ker(\omega_{C,x}).$$

**Proof.** If  $v \in D_x$  but  $v \notin \ker(\omega_{C,x})$ , AER-Max supplies a physical observable  $O = \tilde{O} \circ \pi$  with  $dO_x(v) \neq 0$ . But  $v \in \ker(d\pi_x)$ , so

$$dO_x(v) = d\tilde{O}_{\pi(x)}(d\pi_x(v)) = 0,$$

a contradiction.

**Theorem 29.3 (coisotropic forcing and characteristic identification).** Under AER-Obsmin, AER-Sep, and AER-Max,

$$D = \ker(d\pi) = \ker(\omega_C).$$

Thus the standing surface is coisotropic and admissible redescription coincides with the characteristic foliation of  $\omega_C$ .

**Proof.** Lemma 29.1 gives coisotropy and  $(TC)^{\perp_\omega} \subseteq D$ . For a coisotropic submanifold,  $(TC)^{\perp_\omega} = \ker(\omega_C)$ , so  $\ker(\omega_C) \subseteq D$ . Lemma 29.2 gives  $D \subseteq \ker(\omega_C)$ . Hence equality holds. The characteristic foliation is the foliation tangent to  $\ker(\omega_C)$ , so it is exactly the admissible-redescription foliation.

**Corollary 29.4 (reduced symplectic form).** On the regular quotient there is a unique local symplectic form  $\omega_{phys}$  satisfying

$$\pi^* \omega_{phys} = \omega_C.$$

The descended Poisson bracket is the Poisson bracket induced by  $\omega_{phys}$ .

This theorem is the local mathematical center of the AER specialization. It shows that the gauge/redescription directions of the physical quotient are not appended after the fact. They are forced to be the characteristic directions once observable Hamiltonian evaluation is complete.

### 30. Constraint generators, first-class closure, and multiplier form

Once  $C$  is coisotropic, standard local symplectic geometry supplies local constraint generators. The point is not that the paper assumes Dirac constraint theory. The point is that coisotropy, already forced by the AER interface, entails the local first-class normal form.

**Theorem 30.1 (local constraint generators).** If  $C \subset (S, \omega)$  is coisotropic of codimension  $k$  on a regular stratum, then near each point of  $C$  there are local Darboux coordinates such that

$$C = \{\phi_1 = \cdots = \phi_k = 0\},$$

and the Hamiltonian vector fields  $X_{\phi_a}$  span  $\ker(\omega_C)$  on  $C$ .

This is the standard coisotropic normal form applied after coisotropy has been derived. It is not a hidden premise of the derivation.

**Theorem 30.2 (first-class closure).** The local generators satisfy weak first-class closure:

$$\{\phi_a, \phi_b\} = C_{ab}^c \phi_c$$

for smooth local coefficient functions  $C_{ab}^c$ .

**Proof.** On  $C$ ,  $X_{\phi_a}$  lies in the characteristic distribution and is therefore tangent to  $C$ . Since each  $\phi_b$  vanishes on  $C$ ,  $d\phi_b$  annihilates tangent vectors to  $C$ . Hence

$$\{\phi_a, \phi_b\} = d\phi_b(X_{\phi_a}) = 0$$

on  $C$ . Since the  $\phi_a$  are regular defining functions for  $C$ , Hadamard's lemma gives local smooth functions  $C_{ab}^c$  such that  $\{\phi_a, \phi_b\} = C_{ab}^c \phi_c$ .

The first-class algebra is therefore a consequence of the quotient/characteristic identification. It is not a primitive gauge algebra imposed by hand.

**Definition 30.3 (presentation-parametrized evolution).** AER-P holds when there is a family of Hamiltonians  $H_\alpha$  preserving  $C$  and descending to the same reduced evolution on  $S_{phys}$ , with representative vector-field differences tangent to redescription:

$$(X_{H_\alpha} - X_{H_\beta})(x) \in D_x \quad (x \in C).$$

**Theorem 30.4 (multiplier-type Hamiltonian form).** Under AER-P, each presentation Hamiltonian has the local weak form

$$H_\alpha = \hat{h}_\alpha + u_\alpha^a \phi_a,$$

where  $h_\alpha$  is the reduced Hamiltonian on  $S_{phys}$  and  $\hat{h}_\alpha$  is any local extension of  $\pi^* h_\alpha$  off  $C$ . If the reduced evolution is trivial or the reduced generator is quotiented as dynamically inert, then locally

$$H_\alpha = u_\alpha^a \phi_a$$

modulo weak equality on  $C$ .

**Proof.** Since  $X_{H_\alpha}$  preserves  $C$ , the restriction  $H_\alpha|_C$  is constant along  $D = \ker(\omega_C)$  and hence descends to a function  $h_\alpha$  on  $S_{phys}$ . Choose an extension  $\hat{h}_\alpha$  of  $\pi^* h_\alpha$ . Then  $H_\alpha - \hat{h}_\alpha$  vanishes on  $C$ . Since  $C$  is regularly defined by the  $\phi_a$ , Hadamard's lemma yields  $H_\alpha - \hat{h}_\alpha = u_\alpha^a \phi_a$ . Presentation dependence is thus carried by constraint multipliers rather than by physical quotient content.

**Integrated AER normal form.** The AER specialization can now be stated in one line:

$$\text{AERInterface} + \text{AER-P} \Rightarrow \text{CoisotropicStanding} \wedge \text{CharacteristicRedescription} \wedge \text{FirstClassClosure} \wedge \text{MultiplierForm}.$$

This is the theorem-object bridge from the realized physical interior to GR-type constraint architecture.

### 31. Fixed-domain closure of the AER object

After the local AER object is derived, fixed-domain closure asks whether a non-equivalent same-scope rival can remain. The answer is no, for a specific reason: same-scope physically relevant variation has only two load-bearing loci.

**Definition 31.1 (local AER theorem object).** The derived object is

$$A = (C, \pi : C \rightarrow S_{phys}, D = \ker(\omega_C), \{\cdot, \cdot\}_{phys}, h_{phys}),$$

where  $h_{phys}$  is included when AER-P supplies reduced evolution. A local constraint basis  $\{\phi_a\}$  and multiplier coefficients  $u_\alpha^a$  are presentations of  $A$ , not additional same-scope tensor content.

**Theorem 31.2 (two-locus bridge).** Once AER is derived, same-scope physically relevant structure is exhausted by:

1. quotient physical content on  $S_{phys}$ : physical observables, the descended Poisson bracket, and induced physical continuation;
2. characteristic/redescription structure on  $C$ :  $D = \ker(\omega_C)$  and its local first-class span.

If two presentations agree on these loci, they agree on physical observables, admissible redescription, and same-scope continuation.

**Proof.** Theorem 29.3 identifies redescription with the characteristic foliation. Theorem 30.2 says local first-class generators present that foliation. Theorem 30.4 says presentation Hamiltonians differ from reduced content only by constraint-multiplier terms. Therefore a physically relevant difference can only alter quotient physical content or characteristic/redescription structure. If neither locus changes, the difference is a chart, extension, constraint-basis, multiplier, or notation choice: skin.

**Theorem 31.3 (same-scope AER rival collapse).** Any purported same-scope rival to  $A$  is either lawfully equivalent to  $A$ , skin, hidden load, or a theorem-object change.

**Argument.** A genuine rival must first diverge somewhere. If it changes the standing surface, quotient map, observable algebra, characteristic distribution, or continuation rule, it changes the theorem object. If it changes none of these but later claims physical force, that force must arise by deferred activation, repair, or a second classifier. Those routes violate fixed-domain admissibility. If it never changes either bridge locus, it is skin. Thus no non-equivalent same-scope AER rival remains.

This closure matters for the paper because it prevents the AER specialization from being merely one convenient Hamiltonian representation among many same-scope alternatives. Within the AER comparison class, the constrained-Hamiltonian architecture is the unique same-scope realization of the observable-compatible quotient structure.

## Part VIII - Gravity and Einstein dynamics

The AER theorem object is still more general than gravity. Part VIII uses the gravity-facing specialization and the metric-field-equation projection. The order is crucial:

$$I_D \Rightarrow \text{AER} \Rightarrow \text{AASCGravity} \Rightarrow \Pi_g(A_G) \Rightarrow G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}.$$

The first arrow is role-synthesis; the second is quotient/constraint specialization; the third is role-typing of gravity; the fourth is faithful metric projection; the last is the minimal first-response metric normal form. None of these permits metric dynamics to retro-authorize the interior.

### 32. Gravity as structural necessity

**Definition 32.1 (AASC gravity).** Gravity in the AASC sense is the structural role corresponding to the constraint architecture required for an admissible physical interior: standing surface, admissible redescription, observable quotient, first-class closure, and multiplier-type Hamiltonian presentation.

This definition does not identify gravity with primitive curvature, primitive metric field content, force, or a quantization target. It identifies the role canonically realized by GR-type constrained-Hamiltonian architecture.

**Theorem 32.2 (gravity necessity theorem).** If a system instantiates an admissible physical interior in the regular-stratum AER comparison class and satisfies presentation arbitrariness, then it instantiates gravity in the AASC sense.

**Argument.** The AASC kernel requires admissibility, standing, reference, and irreversibility. Physical standing under redescription requires quotient identity. In regular-stratum Hamiltonian observable evaluation, the AER interface forces coisotropic standing and characteristic redescription. Coisotropic normal form supplies local constraints; first-class closure follows from regularity; presentation arbitrariness supplies multiplier-type Hamiltonian form. These are exactly the clauses of AASC gravity. A system lacking them in the same comparison class does not realize a non-gravitational admissible interior; it fails to instantiate the target.

The word “gravity” is therefore being used in a precise role-theoretic way. A skeptical reader may dislike the terminology, but the theorem does not depend on a semantic trick. It says that the canonical GR constraint role is forced by the admissible physical interior in the AER class. Curvature and metric variables are later geometric presentations of that role.

**Nonclaim boundary.** The theorem does not derive the numerical gravitational coupling, all gravitational dynamics, all possible modified-gravity regimes, global

singular quotient behavior, or a quantum-gravity model. It derives the structural necessity of the GR-type constraint role in the fixed comparison class.

### 33. Canonical GR as downstream realization

Canonical GR supplies the standard geometric realization of the AER/gravity normal form. Let spacetime be foliated by spacelike hypersurfaces  $\Sigma_t$  with ADM data  $(h_{ij}, \pi^{ij})$ . The ADM line element is

$$ds^2 = -N^2 dt^2 + h_{ij}(dx^i + N^i dt)(dx^j + N^j dt),$$

where  $N$  is lapse and  $N^i$  is shift. The canonical Hamiltonian density has the form

$$\mathcal{H}_{can} = N\mathcal{H} + N^i \mathcal{H}_i,$$

with Hamiltonian and momentum constraints

$$\mathcal{H} \approx 0, \quad \mathcal{H}_i \approx 0.$$

The standing surface is

$$C_{ADM} = \{(h, \pi) : \mathcal{H}(x) = 0, \mathcal{H}_i(x) = 0 \text{ for all } x \in \Sigma\}.$$

The smeared constraints satisfy the hypersurface-deformation algebra:

$$\{H[\vec{N}], H[\vec{M}]\} = H[\mathcal{L}_{\vec{N}} \vec{M}],$$

$$\{H[\vec{N}], H[M]\} = H[\mathcal{L}_{\vec{N}} M],$$

$$\{H[N], H[M]\} = H_i[h^{ij}(N\partial_j M - M\partial_j N)].$$

This is first-class closure with structure functions.

**Proposition 33.1 (ADM realization of AER gravity).** On its regular constraint surface, canonical GR realizes the AER gravity normal form: coisotropic standing, characteristic redescription, first-class closure, and multiplier-type presentation dynamics.

**Proof.** The constraint surface  $C_{ADM}$  is the standing surface. The flows generated by the constraints are lawful redescription directions: spatial diffeomorphism and refoliation transformations. The hypersurface-deformation algebra supplies

first-class closure. Lapse and shift are presentation multipliers, so changing them changes representative trajectory while preserving the quotient physical content. Thus ADM is a downstream realization of AASC gravity.

The direction of dependence is important. The paper does not infer AER from ADM. It infers that ADM is the paradigmatic geometric presentation of the AER/gravity role once that role has been forced by admissibility, quotient identity, and observable Hamiltonian closure.

### 34. Metric projection licensing

The gravity role is constraint-theoretic before it is metric. To obtain a field equation one must project the role into a local Lorentzian metric presentation.

**Definition 34.1 (faithful metric projection).** A faithful metric projection is a role-preserving map

$$\Pi_g : A_G \rightarrow (M^4, g_{\mu\nu})$$

such that the metric is downstream from the AASC gravity role, coordinates and frames are skin, curvature and metric response are tensor content, dimension/signature/compatibility conditions are anchors unless separately derived, and no metric equation retro-authorizes the admissibility of the projection.

**Theorem 34.2 (no primitive metric backflow).** No metric description, curvature expression, or field equation can authorize the standing of the physical interior whose projection it inhabits.

**Proof.** A field equation presupposes the metric projection, the metric presupposes comparison and continuation roles, and those roles presuppose admissibility and standing. If the field equation were used to license the interior from which it is projected, a downstream tensor expression would become a boundary selector. That is either hidden anchor import, representative privilege, or same-act repair. All three violate the admissibility kernel. Therefore a metric equation may express projected tensor content but cannot generate the interior that licenses it.

**Minimal compatibility.** In the minimal metric projection the compatible connection satisfies

$$\nabla_\lambda g_{\mu\nu} = 0.$$

Torsion and metric-affine nonmetricity are not declared false. They are classified as richer-scope connection/comparison tensor load when explicitly introduced. In the minimal same-scope metric target, the Levi-Civita comparison envelope is fixed.

### 35. Same-scope first-response metric candidates

The Einstein projection theorem does not classify all possible gravitational theories. It classifies the minimal same-scope first-response metric field-equation target.

**Definition 35.1.**  $\text{SameScopeMetric}_1(E_{\mu\nu})$  holds when  $E_{\mu\nu}$  satisfies all of the following conditions:

1.  $E_{\mu\nu}$  is built only from the licensed metric projection  $(M^4, g, \nabla)$  with minimal metric-compatible connection;
2. it is tensorial under lawful redescription;
3. it is symmetric and covariant rank two;
4. it is local;
5. it is linear in the first admitted curvature obstruction;
6. it contains no curvature products, curvature derivatives, arbitrary curvature functions, torsion, metric-affine nonmetricity, nonlocal operators, boundary/global selectors, extra fields, hidden source content, or representative-dependent terms;
7. it can match a closed same-scope source ledger.

This predicate is not a verbal way of defining Einstein's equation into existence. It fixes the theorem target: minimal, local, metric-internal, first curvature response in a fixed four-dimensional Lorentzian projection. Higher-curvature gravity, scalar-tensor theories, torsion theories, metric-affine theories, nonlocal gravity, semiclassical mixed projections, and horizon endpoint models are not denied. They are typed outside this minimal first-response class unless they collapse to it.

**Lemma 35.2 (curvature-response slots).** If  $\text{SameScopeMetric}_1(E_{\mu\nu})$ , then the only available geometric slots are

$$R_{\mu\nu}, \quad Rg_{\mu\nu}, \quad g_{\mu\nu}.$$

**Argument.** Connection coefficients alone are not tensorial; the first tensorial obstruction of compatible metric transport is curvature. The source-facing equation requires a symmetric rank-two response. The first rank-two contraction of the Riemann curvature is the Ricci tensor  $R_{\mu\nu}$ . The scalar trace  $R$  can return to rank two only by multiplication with the metric, giving  $Rg_{\mu\nu}$ . The zero-curvature homogeneous baseline is  $g_{\mu\nu}$ . Every other candidate either uses curvature products, curvature derivatives, arbitrary functions, extra fields, torsion, metric-affine nonmetricity, nonlocal continuation, boundary/global data, or representative privilege. Those are richer-scope or skin, not minimal same-scope first response.

**Theorem 35.3 (curvature-response normal form).** If  $\text{SameScopeMetric}_1(E_{\mu\nu})$ , then

$$E_{\mu\nu} = aR_{\mu\nu} + bRg_{\mu\nu} + cg_{\mu\nu}$$

for projection-envelope coefficients  $a, b, c$ .

The coefficients are not AMetric defaults or boundary-generated numerical values. Their dimensional interpretation requires source coupling and scale/calibration anchors.

### 36. Einstein field-equation projection

The curvature-response normal form becomes Einstein form only after closed source continuation is imposed.

**Definition 36.1 (closed metric-facing source ledger).**  $T_{\mu\nu}^{closed}$  is the metric-facing tensor ledger of all standing-bearing source roles relevant to metric response within the declared scope: field-carrier mediation, localized particle persistence, deformation-response mass, inertia, bounded transport, interaction/restoration burden, vacuum baseline, and declared boundary/effective contributions. It is closed when no standing-bearing source flux is hidden outside the ledger.

The closed-source condition is

$$\nabla^\mu T_{\mu\nu}^{closed} = 0.$$

This is not imported as a primitive GR conservation law. It is the metric projection of coherent closed source continuation: a source ledger that leaks without a declared exchange channel is incomplete, open/effective, anomalous, or scope-shifted.

**Theorem 36.2 (Einstein-combination theorem).** Let

$$E_{\mu\nu} = aR_{\mu\nu} + bRg_{\mu\nu} + cg_{\mu\nu}.$$

If  $E_{\mu\nu}$  is compatible with a closed same-scope source ledger for general curvature response, then

$$b = -\frac{a}{2}$$

and hence

$$E_{\mu\nu} = aG_{\mu\nu} + cg_{\mu\nu}.$$

**Proof.** Metric compatibility gives  $\nabla^\mu g_{\mu\nu} = 0$ . The contracted Bianchi identity gives

$$\nabla^\mu R_{\mu\nu} = \frac{1}{2} \nabla_\nu R.$$

Therefore

$$\nabla^\mu E_{\mu\nu} = a \nabla^\mu R_{\mu\nu} + b \nabla^\mu (R g_{\mu\nu}) + c \nabla^\mu g_{\mu\nu} = \left( \frac{a}{2} + b \right) \nabla_\nu R.$$

Closed source compatibility requires  $\nabla^\mu E_{\mu\nu} = 0$  for general curvature response, so  $a/2 + b = 0$ . Thus  $b = -a/2$  and the geometric side is  $a G_{\mu\nu} + c g_{\mu\nu}$ .

**Corollary 36.3 (Einstein metric projection).** Under faithful metric projection, `SameScopeMetric1`, and closed source continuation, the minimal first-response metric field equation has the form

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}.$$

**Proof.** Normalize the coefficient  $a$  into the source coupling  $\kappa$  and write  $\Lambda = c/a$ . The homogeneous term  $\Lambda g_{\mu\nu}$  is the quotient-stable metric-facing vacuum ledger, not a primitive zero-point vacuum prediction. The source side is the closed metric-facing ledger of admitted source roles, not primitive mass-energy substance.

**Scope.** This theorem is stronger than compatibility with GR and weaker than derivation of all gravitational physics. It derives the field-equation form in the minimal first-response same-scope metric projection. It does not compute  $\kappa$ , compute  $\Lambda$ , close all matter-sector dynamics, solve the cosmological constant matching problem, or exclude declared richer-scope modifications.

## 37. Modified gravity and high-curvature classification

The Einstein projection theorem classifies alternatives by where they move the theorem target.

**Theorem 37.1 (modified-gravity classification).** Any proposed modification of the Einstein form in the faithful metric projection is exactly one of the following: same-scope collapse to Einstein form, skin, anchor change, hidden tensor load, higher-response curvature load, nonlocal continuation expansion, source-ledger extension, horizon-certificate mechanism, richer-scope theory, or standing failure.

**Argument.** If the candidate satisfies `SameScopeMetric1`, Theorems 35.3 and 36.2 force Einstein form. If it changes no standing-bearing content, it is skin. If it changes dimension, signature, connection type, boundary condition, or locality envelope, it moves an anchor. If it adds curvature products or derivatives, it is higher response. If it adds scalar/vector/tensor fields, torsion, or metric-affine nonmetricity, it declares new tensor or connection load. If it is nonlocal, it expands the continuation envelope. If it uses semiclassical expectation values, it is a mixed projection requiring state and record closure. If it addresses horizon

or high-curvature endpoints, it requires a certificate, leakage/radiation channel, non-sink quotient, or richer scope. If none of these is declared but physical force is claimed, the candidate is hidden load or standing failure.

Representative classifications:

| Candidate                   | Classification                                    |
|-----------------------------|---------------------------------------------------|
| $G_{\mu\nu}$                | same-scope survivor                               |
| $\Lambda g_{\mu\nu}$        | homogeneous vacuum ledger                         |
| $R_{\mu\nu}$ alone          | divergence failure                                |
| $Rg_{\mu\nu}$ alone         | divergence failure                                |
| $R^2, R_{\mu\nu}R^{\mu\nu}$ | higher-response tensor load                       |
| $f(R)$                      | selector/scalar-response theory unless linearized |
| scalar-tensor gravity       | added tensor/source witness                       |
| torsion                     | connection tensor load                            |
| metric-affine nonmetricity  | altered comparison/connection ledger              |
| nonlocal gravity            | continuation-envelope expansion                   |
| semiclassical gravity       | mixed projection requiring record/source closure  |
| horizon endpoint terms      | certificate/leakage/non-sink quotient route       |

This classification does not declare modified gravity false. It prevents modified terms from masquerading as undeclared same-scope first-response alternatives.

## Part IX - Quantum predictive structure and records

Part IX uses the quantum predictive theorem object. The orientation is standard: Hilbert geometry, unitary one-parameter evolution, Bell-type witness sets, and measurement-record structure are familiar quantum objects. The dependency order is not standard-primitive, however. Hilbert space is not used to ground admissibility. It enters as the downstream predictive geometry that survives the AASC quantum-interior constraints, with Stone's theorem and the Jordan-von Neumann reconstruction serving as ordinary mathematical supports inside that companion theorem object.

Part IX uses the quantum side of the standard physics stack. The quantum import has two layers. The interior layer is the Hilbert/Schrodinger predictive theorem object. The boundary layer contains Bell witnesses and measurement-record quotient closure. These are not derived from GR and are not used to derive the RPI role ladder. They are controlled projections of the admissibility-bearing quotient architecture.

### 38. Quantum interior rigidity

**Definition 38.1 (quantum predictive AASC regime).** A quantum predictive regime consists of predictive state spaces for systems, coherent independent composition, nontrivial continuous mixing, admissible updates preserving standing, and no hidden carriers, same-scope generators, repairs, or free continuous tuning not fixed by quotient invariance.

The theorem object is

$$Q_{int} = (\mathcal{H}, \langle \cdot, \cdot \rangle, \{U_t\}_{t \in \mathbb{R}}),$$

with  $U_t = e^{-itH}$  for a self-adjoint generator  $H$  where the usual domain conditions apply.

**Theorem 38.2 (Hilbert/Schrodinger quantum interior).** Under the quantum predictive AASC regime, admissible predictive geometry is Hilbertian and reversible admissible dynamics has Schrodinger form.

**Argument.** Non-Hilbert predictive geometry has to carry its extra standing-bearing content somehow. If it changes subsystem predictions under independent composition without a witness exchange, it is a hidden carrier. If it is selected by a continuous metric, decomposition, or exponent not fixed by invariance, it is free tuning. If it changes standing under admissible updates, it violates standing conservation. If it changes none of the predictive tensor content, it is skin. The surviving representative geometry admits a standing weight. Continuity, independent composition, and isotropic two-dimensional mixing force a quadratic standing weight. The Jordan-von Neumann theorem reconstructs an

inner product from the parallelogram identity. Reversible standing-preserving updates are linear and norm-preserving; strong continuity gives a one-parameter unitary group, and Stone's theorem gives  $U_t = e^{-itH}$ . Thus the predictive interior is Hilbert/Schrodinger up to lawful unitary redescription.

**Nonclaim boundary.** This import does not derive every quantum field theory, all measurement outcomes, all Born-rule interpretation, or a quantum-gravity model. It fixes the predictive interior theorem object used by later Bell and record classifications.

### 39. Bell witness structure

A Bell-scope theorem object fixes operational witness data

$$W_{Bell} = \{P(a, b|x, y), P(a|x), P(b|y)\}$$

with the usual no-signalling consistency conditions included as part of the fixed witness typing. The question is not whether one can enrich the theory with hidden variables, collapse events, or additional state data. The same-scope question is how the fixed Bell witness set is typed when it is used as standing-bearing reusable content.

**Definition 39.1 (joint standing class).** Two internal presentations of a preparation are Bell-standing equivalent when they agree on the full Bell witness set. The equivalence class is the joint standing class of the preparation.

**Theorem 39.2 (Bell tensor-compatibility theorem).** In the nontrivial Bell sector, entanglement is tensor-level compatibility on a joint standing class rather than independently sufficient local standing data or same-scope signalling.

**Argument.** Bell-violating witness sets contain invariant joint constraints not reducible to local factorization on the same operational scope. The joint structure survives lawful redescription and is sampled by local measurement contexts, but a local measurement act cannot create, transmit, repair, or amplify distant standing-bearing tensor content. If it did, it would introduce same-scope standing transfer or hidden causal repair. The fixed witness object instead carries a joint compatibility relation. Local outcomes are admissible local realizations of that already admitted joint class.

**Corollary 39.3 (no same-scope signalling retyping).** A Bell correlation cannot be retyped as hidden same-scope signalling while preserving the same witness object. Any such retyping either changes the theorem object, adds richer state data, or uses a hidden selector.

### 40. Measurement-record quotient

Measurement records are not merely psychological experiences or labels. They are standing-bearing contents used in retained-record reuse, prospective-record

reuse, deliberation, and evidential comparison.

**Definition 40.1 (record-fixation assignment).** Let  $D$  be the set of record-sensitive construction tokens and let  $B \times R$  be physically realized branch-record pairs for a fixed measurement episode. A record-fixation assignment is a map

$$\rho : D \rightarrow B \times R$$

satisfying the fixed quantum scope, physical realization, no-second-selector, and no-repair conditions. Let  $A_{Ref}$  be the space of admissible assignments.

**Definition 40.2 (measurement-record standing equivalence).** Assignments  $\rho$  and  $\sigma$  are measurement-record equivalent, written  $\rho \simeq_{MR} \sigma$ , when they have the same standing-bearing profile across retained-record reuse, prospective-record reuse, deliberation, and evidential comparison. The record quotient is

$$A_{Ref} / \simeq_{MR} .$$

**Theorem 40.3 (measurement-record quotient theorem).** Any standing-bearing presentation of measurement-record content on the fixed quantum scope factors uniquely through  $A_{Ref} / \simeq_{MR}$ .

**Argument.** If two assignments have the same standing-bearing profile, distinguishing them as different theory-level record contents would introduce a second same-scope invariant. If two assignments have different profiles, identifying them would destroy record reference. The quotient by profile equality is therefore both sound and complete for the record-use target. The standard quotient universal property gives unique factorization.

**Theorem 40.4 (selector exhaustion for singular record fixation).** A same-scope package that fixes a singular measurement-record realization does one of four things: changes dynamics, adds ontic privileging, restricts admissible assignments to a single record-equivalence class, or refuses the target explanatory burden. There is no fifth same-scope selector route.

This theorem lets the paper acknowledge collapse, hidden-variable, Everettian, and operational readings without treating any of them as a hidden primitive. If new state data or dynamics are declared, the theory has changed scope. If no such data are declared, the record quotient is the standing-bearing content.

## 41. GR/QM common quotient architecture

The common architecture claim is often easy to overstate. The paper must avoid two false readings:

1. GR and QM are not asserted to be the same mathematical category.

2. ADM phase space is not functorially converted into Hilbert space, nor conversely.

The claim is quotient-architectural.

**Definition 41.1 (common admissibility-bearing quotient architecture).**

A theorem object realizes the common architecture when its same-scope standing-bearing content decomposes into anchors, tensors, and skin; when no second same-scope classifier survives; and when no admissibility-preserving refinement beneath tensor content remains.

**Theorem 41.2 (GR/QM common quotient-architecture theorem, digest).** The AER/GR object and the quantum theorem object realize one common admissibility-bearing quotient architecture, though in different mathematical presentations.

**Argument.** On the GR side, anchors fix the AER/ADM comparison class, tensor content is the observable-compatible quotient plus characteristic/redescription structure, and skin consists of charts, constraint bases, lapse/shift representatives, and extensions that change neither bridge locus. On the quantum side, anchors fix the predictive, Bell, and record domains; tensor content is Hilbert/Schrodinger geometry, Bell joint standing, and measurement-record quotient; skin consists of basis, phase, operator presentation, and branch labels that change no standing-bearing content. Both sides exclude second classifiers, hidden selectors, and fourth same-scope roles by the same fixed-domain discipline. Hence they are projection-equivalent at the level of admissibility-bearing quotient architecture, not identical as formalisms.

**Corollary 41.3 (primitive quantization obstruction).** A primitive same-domain map from the gravitational theorem object into the quantum theorem object carrying independent foundational force does not survive. If it preserves anchors, tensors, quotient identity, and continuation, it factors through the common architecture. If it fails to preserve them, it changes scope or uses hidden load. Effective, semiclassical, loop, string-like, EFT, or other quantum-gravity programs are not rejected; they are typed as restricted-scope or richer-scope constructions requiring certificates.

## Part X - Local Yang-Mills QFT and QFT specialization

The standard reference frame for this part is the Osterwalder-Schrader, Wightman, and Haag-Kastler view of a quantum field theory as a local net or local field theory with a vacuum representation and controlled reconstruction from Euclidean data. The Yang-Mills import is therefore not a verbal claim that fields exist. It is a theorem-object import in the local gauge-invariant sector: the source constructs the local gauge-invariant net generated by flowed curvature composites and records the scope in which a vacuum spectral gap is asserted.

Part X fills the local QFT layer. The paper does not need a theorem for every QFT, but it does need a load-bearing local gauge-QFT theorem object connecting field-carrier, gauge, vacuum, locality, and spectral structure. The companion Yang-Mills package supplies that object for compact connected simple gauge groups in the declared local gauge-invariant scope.

### 42. Local gauge-invariant Yang-Mills QFT theorem object

**Definition 42.1 (Yang-Mills local theorem object).** The companion local Yang-Mills theorem object has the schematic form

$$\text{YM}_{loc} = (\mathcal{A}_{loc}, \Omega, \mathcal{H}, U, \text{OS/Wightman}, \text{Gap}),$$

where  $\mathcal{A}_{loc}$  is a sharp-local gauge-invariant algebra/net,  $\Omega$  is the vacuum vector or vacuum state,  $\mathcal{H}$  is the reconstructed Hilbert space,  $U$  is the spacetime symmetry representation where available, the OS/Wightman entry records reconstruction strength, and **Gap** is the vacuum spectral mass-gap assertion in the declared pure Yang-Mills scope.

The construction route is:

$$\begin{aligned} \text{FlowedCurvComp} &\Rightarrow \text{SharpLocalAlg} \Rightarrow \text{OSRecon} \\ &\Rightarrow \text{WightmanHKNet} \Rightarrow \text{NontrivWitness} \Rightarrow \text{VacuumGap}. \end{aligned}$$

Here the abbreviations denote, respectively, flowed curvature composite approximants, the sharp-local gauge-invariant algebra, Osterwalder-Schrader reconstruction data, Wightman/Haag-Kastler local-net structure, a nontrivial local-field witness, and the vacuum spectral mass-gap statement.

**Theorem 42.2 (local Yang-Mills QFT specialization).** For compact connected simple gauge groups in the declared four-dimensional local pure Yang-Mills scope, the local theorem supplies a local gauge-invariant QFT with sharp-local algebraic structure, OS/Wightman reconstruction strength, nontrivial local field witnesses, and a vacuum spectral mass gap.

**Source note.** The paper uses the Yang-Mills result at the theorem-object level and records its role. The construction does not license fields, gauge, or vacuum from below. Those roles are already part of the RPI ladder. The Yang-Mills theorem shows that the field/gauge/vacuum layer has a mature local QFT realization in a central pure-gauge case.

**Scope guard.** The import is not all QFT. It does not supply a constructive interacting Standard Model, all matter-field sectors, all scattering theory, all global/topological line and surface sectors, all non-simple groups, all EFT/RG regimes, or all curved-background QFT. Global-form-sensitive sectors are typed as additional/global data rather than as part of the sharp-local net.

**Source note.** Exact Yang-Mills theorem packets for flowed curvature composites, sharp-local algebras, OS/Wightman reconstruction, nontriviality, and the gap statement are listed in Appendix A.

### 43. QFT as local field-carrier specialization of the RPI

The Yang-Mills theorem object gives the paper a precise QFT bridge:

$$(LCD_D, \Phi_D, G_D, \Omega_D, \text{BoundTrans}_D) \Rightarrow \text{YM}_{loc}$$

as a theorem-object specialization. The arrow records a controlled attachment of a mature local QFT object to the RPI field/gauge/vacuum roles, not a primitive source for those roles.

The role map is:

| QFT/YM structure                     | RPI role                                       |
|--------------------------------------|------------------------------------------------|
| local net $\mathcal{A}(\mathcal{O})$ | local continuation domains plus field carriers |
| flowed curvature composites          | gauge-field carrier witnesses                  |
| sharp-local gauge-invariant algebra  | $G_D$ -compatible tensor content               |
| vacuum vector/state                  | $\Omega_D$                                     |
| Wightman/OS reconstruction           | predictive theorem-object representation       |
| nontrivial local field witness       | non-skin field-carrier content                 |
| spectral mass gap                    | vacuum-sector tensor content                   |
| global-form line/surface sectors     | declared global/richer-scope data              |

**Proposition 43.1 (RPI-to-QFT typing).** The local Yang-Mills theorem object specializes the field, gauge, vacuum, locality, and transport roles of  $I_D$  without turning gauge-variant presentation into physical multiplicity.

**Proof.** The local net assigns standing-bearing observable content to local regions, matching the local-domain role. Gauge-invariant local composites provide field-

carrier witnesses, not gauge-variant skin. The vacuum representation supplies a non-null baseline sector. The mass gap is a spectral property of that vacuum representation. Since the local algebra is gauge-invariant, presentation-equivalent gauge variants do not become distinct physical occupants. Thus the theorem object realizes the RPI field/gauge/vacuum roles at QFT strength.

#### 44. Scope guard for QFT specialization

A skeptical reader may object that Yang-Mills is not all QFT. That objection is correct but not damaging. The import is not advertised as all QFT.

**Theorem 44.1 (QFT import scope).** The Yang-Mills import licenses a local gauge-invariant pure-gauge QFT theorem object, not a universal theorem for every quantum field theory.

**Proof of scope classification.** Matter QFTs add representation, Yukawa, mass, and source ledgers. The Standard Model structural tuple supplies those ledgers structurally but not as a full constructive QFT. Scattering theory adds asymptotic and detector records. EFT/RG adds scale-dependent effective-scope transport. Curved-spacetime QFT adds background and local covariance ledgers. None of these is negated by the Yang-Mills import; they are simply not the theorem target. The import closes the local pure-gauge QFT layer needed by the paper.

The QFT stack is therefore:

1. RPI derives fields, gauge, vacuum, and local domains as roles.
2. The Yang-Mills theorem object supplies a sharp local gauge-invariant QFT realization of the pure-gauge sector.
3. The ZPE/renormalization bridge supplies QFT quotient discipline for homogeneous vacuum offsets.
4. The SM structural theorem supplies matter/gauge/Higgs/Yukawa/anomaly/vacuum structural specialization.

Together these are sufficient for a physics-stack synthesis without pretending to derive every QFT in the main paper.

## Part XI - Vacuum, renormalization, and cosmological constant discipline

The standard physics frame here is renormalized QFT and semiclassical gravity. The stress tensor is a composite operator; its homogeneous metric-proportional component can be shifted by finite local counterterm freedom, and in semiclassical gravity that shift is not separately identifiable from a compensating shift of the cosmological term. The AASC contribution is not a new vacuum dynamics. It is the quotient statement that the same finite-renormalization invariance is exactly standing preservation in the restricted homogeneous sector.

The vacuum role  $\Omega_D$  says that no-particle is not nullity. The QFT vacuum-renormalization uses explain how scalar vacuum-energy assignments acquire or fail standing in standard QFT practice. This part prevents vacuum ontology, zero-point summation, and cosmological-constant overclaims from contaminating the RPI stack.

### 45. QFT quotient condition and standing preservation

Consider the homogeneous metric-proportional vacuum sector

$$\langle T_{\mu\nu} \rangle_{0,R} = -\rho_R g_{\mu\nu}.$$

Finite renormalization permits

$$\rho_R \mapsto \rho_R + \delta\rho.$$

In semiclassical gravity, this is accompanied by

$$\Lambda_R \mapsto \Lambda_R - 8\pi G_R \delta\rho,$$

leaving

$$\Lambda_{eff} = \Lambda_R + 8\pi G_R \rho_R$$

fixed.

**Definition 45.1 (renormalization quotient in the homogeneous sector).**

Two descriptions of the homogeneous vacuum sector are equivalent when they differ by finite metric-proportional stress-tensor shifts and associated convention-preserving transformations that preserve the empirical witness ledger. A scalar has standalone predictive content only if it descends to this quotient.

**Definition 45.2 (standing preservation in the same sector).** A scalar has AASC standing only if it is invariant under admissible redesciptions preserving

the witness ledger: transition frequencies, scattering data, response quantities, phase differences, Casimir-type forces, and curvature records after matching.

**Theorem 45.3 (renormalization-standing equivalence).** In the restricted homogeneous vacuum sector, QFT renormalization-quotient invariance and AASC standing preservation are the same quotient condition.

**Proof.** The admissible transformations on the AASC side are exactly the finite renormalization and convention-preserving transformations on the QFT side in this sector. If a scalar has standing, it is invariant under all such witness-preserving transformations and therefore descends to the QFT quotient. If it descends to the QFT quotient, it is stable under the admissible transformations preserving the witness ledger and therefore has standing. Thus the two filters select the same scalar assignments. The equivalence is local to the homogeneous metric-proportional sector; it does not classify the full stress tensor in arbitrary curved backgrounds.

**Corollary 45.4.** The absolute zero-point coefficient fails both tests for the same reason: it is a global additive offset without invariant relational witness.

## 46. No standalone zero-point cosmological prediction

For a relativistic field, the formal zero-point expression is schematically

$$\rho_{zp}(\Lambda_{UV}) = \frac{1}{2} \int^{\Lambda_{UV}} \frac{d^3k}{(2\pi)^3} \sqrt{k^2 + m^2}.$$

The questionable step is not computing a regulated contribution. The questionable step is treating its absolute value as a standalone cosmological prediction.

**Theorem 46.1 (no standalone absolute zero-point prediction).** No nonconstant scalar depending only on  $\rho_R$  in the homogeneous metric-proportional sector is invariant under  $\rho_R \mapsto \rho_R + \delta\rho$ . Therefore the absolute homogeneous matter-sector vacuum energy is not a standalone QFT prediction.

**Proof.** If  $X(\rho_R)$  is invariant under all finite translations, then

$$X(\rho) = X(\rho + \delta\rho)$$

for every finite  $\delta\rho$ . Translation orbits are the entire real line, so  $X$  is constant on the quotient. A constant carries no information about the absolute value of  $\rho_R$ . Thus no nonconstant absolute zero-point scalar descends to the predictive quotient.

**Theorem 46.2 (three-role exhaustion for vacuum-energy-like quantities).** In the restricted target, every admissible vacuum-energy-like scalar is one of:

1. skin: an absolute offset changed by witness-preserving redescription;
2. relational standing content: differences, derivatives, transitions, phase comparisons, Casimir-type responses, or boundary-conditioned quantities;
3. boundary-only input: an empirically fixed cross-envelope parameter such as the matched  $\Lambda_{eff}$ .

There is no fourth role in which the absolute zero-point energy of a single fixed vacuum sector is an independent standing-bearing prediction.

**Consequences.** Casimir forces survive because they are relational response quantities. Phase-transition and metastable-vacuum energy differences survive because they compare dynamically related configurations. The observed effective cosmological constant remains physical and unresolved as a matching/naturalness problem. What is removed is only the prediction-forming procedure that compares a regulator-dependent zero-point sum with the measured cosmological parameter as though the former had already become a QFT prediction.

## 47. Vacuum ledger in Einstein projection

The Einstein projection theorem uses the homogeneous term

$$\Lambda g_{\mu\nu}$$

as metric-facing tensor content. The ZPE discipline says how to read it.

**Proposition 47.1 (cosmological term as quotient-stable homogeneous ledger).** In the faithful metric projection,  $\Lambda g_{\mu\nu}$  is admissible as homogeneous quotient-stable metric-facing vacuum ledger content, not as primitive zero-point vacuum energy.

**Proof.** The RPI vacuum role supplies a non-null no-particle baseline. The Einstein projection permits the homogeneous metric baseline  $g_{\mu\nu}$  with coefficient  $\Lambda$ . The ZPE quotient discipline blocks the split between matter-sector vacuum coefficient and cosmological counterterm from carrying standalone standing. Therefore the standing-bearing object is the effective homogeneous metric ledger, not a cutoff zero-point prediction.

This is the place where QFT practice and AASC role discipline align. The vacuum is physically real as a baseline role; relational quantum vacuum effects are physical; the effective cosmological term is physical. But absolute homogeneous zero-point offset is not a standalone prediction-forming object.

## Part XII - Standard Model structural specialization

The Standard Model import is structural rather than Lagrangian-first. The conventional product group, chiral representation table, Higgs doublet notation, Yukawa operators, charges, and anomaly ledger are displayed because they are the familiar physics-facing presentation of the recovered structural occupant. They are not treated as primitive data from which AASC roles are reverse-engineered.

Part XII uses the Standard Model structural theorem object. With this source in the stack, the Standard Model is no longer merely a correspondence target. It is a downstream structural specialization of the realized physical interior.

### 48. Standard Model structural specialization

The SM structural source is dependency-compressed. It starts from the AASC kernel, AMetric/interface discipline, realized physical interior, gauge/role carriers, mass-response organization, and support modules for gauge-product recovery, representation-response rigidity, charge-lattice rigidity, anomaly neutrality, Higgs/bridge fixation, Yukawa-realizability, vacuum compatibility, and parameter-status discipline.

**Theorem 48.1 (SM structural specialization).** Given the AASC kernel, interface layer, realized physical interior, gauge-realization ladder, PSM/SR survivor modules, Higgs/bridge support, Yukawa-realizability support, charge/anomaly/vacuum compatibility, and mass-response slots, the recovered standalone Standard Model structural tuple is fixed up to admissible redescription.

This import is structural. It does not assert closure of the complete empirical parameter table. It does not use the accepted Standard Model Lagrangian as a primitive premise in the paper. It uses the result that the conventional Standard Model structural occupant is the survivor of the AASC gauge/role and representation-response rigidity chain.

The compressed support chain is:

$$I_D \Rightarrow G_D \Rightarrow \text{GaugeRealization} \Rightarrow \text{PSM/SR} \Rightarrow \Sigma_{SM}/ \simeq_{Adm} .$$

The relevant gauge-realization ladder is:

$$G_D \Rightarrow Gpd_{\partial}(D) \Rightarrow \nabla_{\partial D} \Rightarrow \Omega_{\partial D} \Rightarrow \text{Rep/Charge} \Rightarrow A_{anom} = 0 \Rightarrow \text{GaugeVac}_D \Rightarrow G_{SM}.$$

Only after these role and survivor constraints are composed is the conventional presentation  $SU(3)_c \times SU(2)_L \times U(1)_Y$  displayed.

**Source note.** The gauge-realization, PSM/SR, Higgs/bridge, Yukawa, charge, anomaly, vacuum, and parameter-status source labels are collected in Appendix A. The main text records the structural chain and the recovered tuple.

## 49. Recovered Standard Model tuple

The recovered structural tuple is

$$\Sigma_{SM} = (G_{SM}, R_{SM}, H_{br/fix}, O_Y, Q_{SM}, A_{anom}, V_{SM}, M_{slot}, \Theta_{mix}, N_g) / \simeq_{Adm} .$$

Here

$$G_{SM} = SU(3)_c \times SU(2)_L \times U(1)_Y.$$

The entries have the following meanings:

- $G_{SM}$  is the recovered gauge-product survivor.
- $R_{SM}$  is the chiral representation-response skeleton.
- $H_{br/fix}$  is the Higgs-like bridge/fixation carrier.
- $O_Y$  is the Yukawa-realizability and coupling-slot ledger, not the numerical Yukawa matrix table.
- $Q_{SM}$  is the rigid charge-lattice ledger.
- $A_{anom} = 0$  is the anomaly-neutral survivor ledger.
- $V_{SM}$  is the vacuum-compatible residual normal form.
- $M_{slot}$  records mass-response slots.
- $\Theta_{mix}$  records mixing/phase slots.
- $N_g = 3$  is the family-cardinality envelope record of the standalone SM structure.

The conventional field-role presentation of the matter skeleton is:

| Field role | Conventional representation | Structural status                       |
|------------|-----------------------------|-----------------------------------------|
| $Q_L$      | $(3, 2)_{1/6}$              | left-quark doublet<br>response slot     |
| $u_R$      | $(3, 1)_{2/3}$              | up-type singlet response<br>slot        |
| $d_R$      | $(3, 1)_{-1/3}$             | down-type singlet<br>response slot      |
| $L_L$      | $(1, 2)_{-1/2}$             | lepton-doublet response<br>slot         |
| $e_R$      | $(1, 1)_{-1}$               | charged-lepton singlet<br>response slot |

| Field role   | Conventional representation   | Structural status                                                        |
|--------------|-------------------------------|--------------------------------------------------------------------------|
| $H_{br/fix}$ | conventionally $(1, 2)_{1/2}$ | bridge/fixation carrier;<br>scalar-sector numerical<br>values downstream |

This display is included to block a common misreading. The SM import is not merely a role schema. The role language is the derivation medium; the recovered object is the explicit structural tuple with conventional gauge and representation presentation.

## 50. Gauge product and representation-response structure

**Theorem 50.1 (gauge-product survivor).** Within the declared SM structural envelope, the gauge-product survivor has the conventional presentation

$$SU(3)_c \times SU(2)_L \times U(1)_Y$$

up to admissible redescription and normalization conventions.

**Argument.** The physical interior first supplies fields, particles, transport, and boundary-fixed gauge equivalence as roles. The gauge-realization sequence then forces transition organization, connection-like standing-preserving transport, curvature/holonomy obstruction, representation admissibility, charge-witness rigidity, anomaly obstruction, and gauge-vacuum compatibility. PSM/SR modules close the product-sector, representation-response, Abelian charge, anomaly, and residual/vacuum survivor ledgers. The conventional Standard Model gauge product is the surviving physics-facing presentation of that ledger. Extra gauge factors are same-scope rivals only if they preserve all representation, charge, anomaly, witness, vacuum, transport, and boundary ledgers. If they do, they are quotient-redundant; if they do not, they declare larger-domain load.

**Theorem 50.2 (representation-response skeleton).** The chiral representation-response skeleton is structural rather than parameter-level.

**Argument.** Changing the representation skeleton changes operator availability, anomaly accounting, charge response, Yukawa compatibility, boundary book-keeping, and vacuum compatibility. It therefore changes tensor content, not a numerical value inside a fixed object. Thus  $R_{SM}$  belongs to the structural tuple. By contrast, numerical gauge couplings and fitted Yukawa eigenvalues vary inside the structural envelope and require downstream parameter modules.

## 51. Higgs bridge and Yukawa-realizability ledger

The SM source separates Higgs-like bridge/fixation from Higgs-sector numerical phenomenology.

**Definition 51.1 (Higgs-like bridge/fixation carrier).**  $H_{br/fix}$  is the structurally required carrier mediating admitted order standing into admitted deformation and mass-response standing over the recovered SM skeleton. In the support notation, its carrier is represented by a bridge quotient such as

$$B_D = (W_D^{br} / \sim_D^{br}, \text{Sig}^{br}, \beta_D^{red}, \rho^{br}).$$

It is not a raw scalar coordinate, vev value, potential coefficient, normalization convention, empirical weak scale, Yukawa term, mass matrix, or mixing datum.

**Theorem 51.2 (Higgs/bridge structural status).** Once gauge-invariant SM bookkeeping closes while stable mass parameters are required, a Higgs-like bridge/fixation carrier is structurally unavoidable in the declared SM envelope.

**Argument.** Stable mass-response standing must be mediated between representation/gauge order and deformation/inertial response. A hidden fixation structure would fail standing because it would carry mass-response load without an admitted witness. A bridge/fixation carrier supplies the required operator-compatibility and deformation-response route. The theorem does not derive the Higgs potential, vev, scalar mass, self-coupling, weak mixing angle, gauge-boson masses, or fermion mass values.

**Definition 51.3 (Yukawa-realizability ledger).**  $O_Y$  records structural operator availability and coupling-witness topology over the bridge carrier. In the charged sectors it records the up-type, down-type, and charged-lepton Yukawa slots. In the neutrino sector it records an explicit status variable rather than silently choosing Dirac, Majorana, Weinberg, sterile, or absent structure.

**Theorem 51.4 (Yukawa realizability as structural operator availability).** Under the minimal chiral load with the conventional one-doublet bridge skin, anomaly freedom and complete Yukawa-operator realizability are equivalent tensor requirements across the unbroken chiral-gauge and broken-phase bookkeeping skins.

**Argument.** If Yukawa operators are not realizable, transport to the broken-phase skin leaves non-transportable massless chiral remnants or requires compensating load not present in the fixed structural object. If anomaly cancellation holds while Yukawa realizability fails, the two descriptions do not carry the same standing-bearing load under phase transport. Conversely, complete Yukawa realizability without anomaly-compatible standing would carry operator availability on an inadmissible representation ledger. Thus  $O_Y$  is structural. Numerical Yukawa matrices remain downstream.

## 52. Parameter-status discipline

The SM structural theorem distinguishes object structure from numerical instantiation.

**Structural entries.** Gauge product, representation-response skeleton, Higgs bridge/fixation carrier, Yukawa-operator availability, charge lattice, anomaly-neutral ledger, vacuum-compatible residual form, mass-response slots, mixing/phase slots, and  $N_g = 3$  belong to  $\Sigma_{SM}$ .

**Parameter-level targets.** Numerical gauge couplings, Yukawa eigenvalues, quark masses, charged-lepton pole masses, CKM/PMNS numerical coordinates, CP phase values, Higgs potential, vev, scalar mass, self-coupling, weak-scale values, and absolute scale anchors remain downstream unless a separate module supplies a witness, exactness certificate, quotient collapse, reconstruction rule, or zero-anchor derivation.

**Proposition 52.1 (structure/parameter separation).** A datum belongs to the structural tuple when changing it changes the admissible object class, role ledger, witness family, operator-availability ledger, transport ledger, anomaly ledger, or vacuum survivor ledger. A datum belongs to the parameter program when it changes a value inside the fixed structural occupant without changing those ledgers.

This distinction is essential for referee control. Failure to derive  $g_1, g_2, g_3$ , all Yukawa values, all masses, CKM/PMNS values, or the Higgs potential is not failure to derive the standalone SM structure. It is an open parameter-sector problem.

### 53. Charged-lepton parameter-sector module

The charged-lepton arc is included only as the first downstream parameter-sector module. It is not a premise of SM structural closure.

The module isolates a charged-lepton role-cell residue

$$\delta_\ell = \frac{2}{9}$$

and a square-root amplitude carrier with a  $Z_3$ -circulant structure. A one-anchor metric display uses the electron mass as a scale skin import and produces numerical muon and tau readouts as diagnostics. Subsequent protected pole-boundary analysis narrows the residual but does not close it exactly.

**Status theorem 53.1 (charged-lepton module).** The charged-lepton module provides a conditional one-anchor numerical readout and protected residual localization. Exact pole closure and zero-anchor absolute-scale derivation remain open.

**Source note.** The construction uses the structural SM tuple and mass-response slots, plus numerical-readiness and charged-lepton support modules. Observed muon and tau masses are not construction inputs. The electron mass is a one-anchor metric skin import. The remaining protected trace remainder is

explicitly open. Therefore the module demonstrates that the parameter program is nontrivial without pretending that the full SM parameter table is closed.

## Part XIII - Unified standard-physics stack

### 54. Standard physics stack typing proposition

The preceding parts do not show that every standard theory is generated directly from the RPI tuple in one mathematical operation. They show a typed dependency stack. The appropriate statement is therefore a stack-typing proposition, not a new derivation theorem competing with the companion sources.

**Proposition 54.1 (standard physics stack typing).** Once the realized physical interior  $I_D$  is fixed, the mature physics theorem objects used in this paper occupy the following downstream positions:

$$I_D \rightsquigarrow \{\text{AER}, \text{AASCGravity}, \text{EinsteinProjection}, \text{QuantumInterior}, \text{YMLocalQFT}, \text{QFTRenStanding}, \Sigma_{SM}\}.$$

The symbol  $\rightsquigarrow$  denotes typed projection or specialization. It does not denote a single uniform derivation relation.

**Proof.** The AER object is the quotient/constraint specialization of admissible physical standing in regular-stratum Hamiltonian observable evaluation. AASC gravity is the constraint role of that admissible physical interior when standing surface, quotient observables, redescription, first-class closure, and multiplier presentation are fixed. Einstein projection is the faithful Lorentzian metric projection of the gravity role under the minimal first-response same-scope predicate. Quantum interior is the predictive projection fixing Hilbert geometry and Schrodinger-form reversible dynamics. The Yang-Mills object is the local gauge-field theorem-object specialization of the field/gauge/vacuum layer. The ZPE result is the QFT renormalization-standing bridge in the homogeneous vacuum sector. The Standard Model tuple is the structural gauge-field-particle-mass-vacuum specialization after gauge-realization, PSM/SR, Higgs/bridge, Yukawa-realizability, charge, anomaly, vacuum, and parameter-status modules are composed.

Each entry is downstream of  $I_D$  in the relevant type. None is used as a primitive source for the RPI ladder. Conversely, omitting these entries would leave the paper isolated from the central standard-physics theorem objects. The proposition therefore records the typed stack of mature projections and specializations.

This stack statement blocks two opposed errors. The first error is to say that AASC ignores standard physics. The companion theorem objects show otherwise. The second error is to say that AASC assumes standard physics as a primitive foundation. The no-backflow rule excludes that reading.

### 55. Claim-status matrix for the standard physics stack

A full claim-status matrix accompanies the paper as `ledgers/claim_status_matrix.csv`; this section gives the publication-facing summary. Each major physics object is

assigned a type, source class, scope, and nonclaim boundary. That discipline is part of the argument, because the paper uses a large theorem stack and cannot allow source roles to drift.

RPI is the role-level synthesis theorem of the first half of the paper. AER supplies the quotient/constraint theorem object. AASC gravity supplies the role-theoretic constraint theorem. The Einstein field-equation form supplies the minimal metric projection. Hilbert/Schrodinger quantum structure supplies predictive rigidity. Bell and measurement-record closure supply boundary classifications. Yang-Mills QFT supplies the local gauge-invariant theorem object. ZPE supplies the renormalization-standing bridge. The Standard Model tuple supplies the structural specialization. The full parameter table, all QFTs, all cosmology, and full quantum gravity remain outside the claim set of this paper.

This matrix is not editorial apparatus. It prevents equivocation between a theorem object, a correspondence, a projection, a specialization, and a parameter target. Ambiguous claim status would itself be a route for hidden load, overreporting, or backflow.

## 56. Standard-physics correspondence table

The familiar structures of physics are located in the realized physical interior as role specializations. Spacetime represents local-domain, comparison, continuity, order, and causal roles. The metric tensor specializes quotient-stable comparison, transport reach, and faithful metric projection. Fields specialize  $\Phi_D$ . Particles and excitations specialize  $\Pi_D$ . Gauge groups, connections, curvature, and holonomy specialize  $G_D$  and transport obstruction. Mass terms and inertial parameters specialize  $M_D$ . Vacuum states and sectors specialize  $\Omega_D$ . Propagators and commutator support specialize bounded transport and causal dependence. Dynamics specializes continuation, transport, and deformation response. Hilbert space specializes predictive geometry. Measurement records specialize record-fixation quotients. The Standard Model specializes the gauge-field-particle-mass-vacuum structural tuple.

This correspondence table does not reduce physics to terminology. It identifies where later mathematical structures perform the roles already forced by standing-bearing physical persistence. If a proposed correspondence changes role content, it is no longer mere correspondence; it is tensor load or scope change.

## Part XIV - Route exhaustion

### 57. Primitive-source route exhaustion

The realized physical interior also has a negative theorem: primitive-source attempts do not survive as foundations for the same target. Primitive spacetime, primitive metric, primitive matter, primitive field, primitive particle, primitive gauge group, primitive mass scalar, primitive vacuum, primitive time, primitive causality, primitive information, primitive GR, primitive QM, primitive QFT, primitive Standard Model, formal-only models, and empirical-fit-first models each attempt to supply physical interiority from a favored structure.

Each route has one of two outcomes. Either it supplies all the roles required by  $I_D$ , in which case it has become a representation or specialization of the realized physical interior; or it omits some required role, uses hidden selectors, changes scope, collapses to skin, or repairs standing failure after the fact. The mature-theory routes require special care. GR, QM, QFT, Yang-Mills, and the Standard Model are not false and not dispensable. They fail only as primitive sources. Their proper status is downstream theorem object, projection, or structural specialization.

A same-scope counterexample must preserve the target of standing-bearing physical interiority while avoiding  $I_D$  and all listed failure modes. Merely pointing to a successful mature theory is not enough, because successful mature theories already instantiate the relevant role architecture.

### 58. Same-scope rival normal forms

The finite normal-form family for same-scope rivals consists of anchor motion, tensor-load import, quotient motion, continuation motion, hidden selector, repair or deferred activation, representative privilege, metric backflow, parameter laundering, and skin-only formalism.

The collapse pattern is uniform. Anchor motion changes the theorem object. Tensor-load import declares richer scope. Quotient motion changes physical identity. Continuation motion changes dynamics. A hidden selector violates the no-second-classifier rule. Repair or deferred activation violates irreversibility. Representative privilege promotes skin to tensor. Metric backflow uses a downstream projection to authorize its own licensing conditions. Parameter laundering converts a fitted or calibrated value into an illicit structural premise. Skin-only formalism changes no standing-bearing content.

The value of the normal-form theorem is that objections become typed. A critic need not accept the vocabulary of the paper to challenge it; they need only supply a same-scope physically relevant difference outside the family. The family therefore functions as the final test of minimality and closure.

## Part XV - Objections and conclusion

### 59. Objections and replies

The preceding route-exhaustion results make objections type-checkable. The most important objections are answered below in the same dependency order used by the paper.

**Terminology objection.** The paper is not protected by vocabulary. Each role carries a deletion test: remove it and a specific physical operation fails. Locality, comparison, transport, field-carrier mediation, particle identity, gauge equivalence, deformation response, vacuum baseline, temporal order, and causal dependence are load-bearing roles, not alternate names for already assumed objects.

**Circularity objection.** The kernel does not define the conclusion into existence. Raw sequences are distinguished from governed constructions, and target adequacy forces reference, standing, irreversibility, and admissibility before physical roles are introduced. The physical ladder is downstream of that target adequacy result.

**Metric-smuggling objection.** Metricity is not assumed at the boundary. The AMetric discipline blocks primitive metric, coordinate, ordering, probability, scale, or comparison authority. Metric role first enters as quotient-stable comparison and later as faithful Lorentzian projection.

**GR objection.** GR is not used to derive AER or the realized physical interior. AER is developed as the quotient/constraint theorem object; ADM is then read as the paradigmatic downstream realization of that structure. Einstein dynamics enters only at the faithful metric-projection level.

**Einstein/Lovelock objection.** The Einstein projection is not a bare appeal to a metric uniqueness theorem. Its target is the minimal same-scope first-response metric projection after AASC gravity, metric-licensing, closed source continuation, and ATS role discipline. Modified-gravity terms are not declared false; they are typed as higher response, richer scope, nonlocal continuation, extra tensor load, or certificate regimes.

**Quantum objection.** Hilbert space is not a primitive foundation in the proof. It is the companion predictive theorem object. Bell witnesses and measurement records are boundary-level quotient structures. The GR/QM common architecture claim is role-preserving quotient-architecture equivalence, not category identity between ADM phase space and Hilbert/operator formalisms.

**QFT objection.** The paper does not claim to derive every QFT. It uses a local gauge-invariant Yang-Mills theorem object in a precise pure-gauge scope and combines it with the RPI field/gauge/vacuum roles and ZPE renormalization-standing bridge. That is enough for the local gauge-QFT layer; matter QFTs, scattering programs, and full SM constructive QFT remain separate targets.

**Standard Model objection.** The Standard Model is not reverse-engineered from a starting Lagrangian. The companion SM source gives a structural tuple after the AASC kernel, interface discipline, realized physical interior, gauge-realization ladder, PSM/SR survivor structure, Higgs/bridge support, Yukawa-realizability support, charge/anomaly/vacuum compatibility, and parameter-status modules. The full empirical parameter table is explicitly not claimed closed.

**Vacuum-energy objection.** The paper does not deny vacuum physics. It denies only the prediction-forming route that treats a regulator-dependent absolute zero-point offset as a standalone cosmological prediction. Casimir forces, phase differences, response quantities, and the effective cosmological term survive because they are relational or matched quotient-stable quantities.

**Quantum-gravity objection.** Effective, semiclassical, loop, string-like, EFT, and other quantum-gravity programs are not rejected. A primitive same-domain map from GR to QM is typed as ill-formed; a constraint-native richer-scope realization with explicit anchors, tensors, records, and transport certificates remains a legitimate future program.

These replies do not ask the reader to accept every companion theorem uncritically. They state the burden precisely. A same-scope counterexample must preserve the target, avoid the normal-form failure family, and carry physical standing without backflow, hidden selectors, or scope change.

## 60. Conclusion

The central result is that the realized physical interior is neither a primitive arena nor an inventory of physical things. It is the minimal admissible role architecture in which standing-bearing physical persistence can be non-null, localized, locally continued, comparatively represented, transported, carrier-mediated, particle-identifiable, gauge-stable, deformation-responsive, vacuum-supported, temporally ordered, and causally dependent.

The first half of the paper establishes this at the role level. The AASC kernel fixes the constructional conditions under which determinate same-domain content can stand: admissibility, standing, reference, and irreversibility. The AMetric and anchor/tensor/skin disciplines prevent metric, coordinate, scale, field, probability, information, or presentation choices from entering as primitive standing sources. The realized-physics ladder then forces the physical roles one by one. Non-vacuous persistence blocks null physical realization. Bounded standing-localization supplies the first locality role. Continuation loci and local domains organize local physical persistence. Quotient-stable comparison supplies the metric role without primitive metricity. Bounded perturbation transport supplies finite-propagation discipline. Fields become distributed persistence carriers; particles become localized persistence classes; gauge becomes boundary-fixed presentation equivalence; mass and inertia become deformation-response profiles; vacuum becomes the non-null no-particle baseline; time becomes irreversible

continuation ordering; causality becomes witness-dependent dependence across that order.

The main RPI theorem packages those roles as

$$I_D = (\text{NonVacRec}_D, [B_D]_{\equiv B}, \text{Adj}_D, N_D, \text{LCD}_D, \text{Cmp}_D, \text{BoundTrans}_D, \Phi_D, \Pi_D, G_D, M_D, \Omega_D, T_D, \text{Cau}_D).$$

The theorem says that this package satisfies coherence, closure, and operational minimality. Coherence means that the roles preserve one standing-bearing physical content rather than forming a catalogue. Closure means that the required standing-bearing operations are handled without null collapse, hidden selector import, primitive container ontology, primitive matter inventory, primitive metric backflow, pathless transport, gauge artifact, vacuum-substrate confusion, primitive time import, or skin-only formalism. Operational minimality means that deleting any major role family breaks the physical interior at the role level.

The second half of the paper shows that this interior is not isolated from standard physics. It develops mature theorem objects at controlled type. The AER quotient/constraint theorem specializes the admissible physical interior to regular-stratum Hamiltonian observable evaluation and forces coisotropic standing, characteristic redescription, first-class closure, and multiplier-type Hamiltonian form. Gravity in the AASC sense is then the forced constraint role of the admissible physical interior. Canonical GR realizes that role through ADM constraint structure, gauge/redescription, and lapse/shift presentation freedom. A faithful Lorentzian metric projection of the gravity role yields the Einstein field-equation form

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}$$

as the unique minimal first-response same-scope metric normal form under closed source continuation. This is not a derivation of all gravitational physics, all modified-gravity exclusions, or numerical values for  $\kappa$  and  $\Lambda$ . It is a theorem about the minimal metric projection of the AASC gravity role.

Quantum mechanics enters through a different projection. The quantum predictive theorem object forces Hilbertian inner-product geometry and Schrodinger-form reversible dynamics under the relevant AASC predictive constraints. Bell witnesses and measurement records are then boundary-level quotient structures: entanglement is tensor-level compatibility on a joint standing class, and measurement-record content factors through the measurement-record quotient. The GR/QM common architecture theorem does not identify ADM phase space with Hilbert space. It says that both theorem objects factor through the same admissibility-bearing quotient architecture: anchors fix well-typedness, tensors carry standing-bearing content, and skin carries presentation surplus.

The local QFT layer is represented by the Yang-Mills theorem object. It gives a

local gauge-invariant pure Yang-Mills QFT realization with sharp-local algebraic structure, OS/Wightman reconstruction strength, nontrivial gauge-invariant local field witnesses, and a vacuum mass gap in the declared compact connected simple-group scope. This does not derive every QFT, the full interacting Standard Model as a constructive QFT, all scattering theory, or all global sectors. It supplies the load-bearing local gauge-QFT theorem object needed by the paper.

The vacuum-renormalization layer prevents a common overclaim. In the homogeneous metric-proportional vacuum sector, AASC standing preservation and QFT renormalization-quotient invariance are the same quotient condition. The absolute zero-point coefficient is a global additive offset without invariant relational witness. It is not a standalone cosmological prediction of QFT. Casimir forces, phase differences, and response quantities survive because they are relational. The effective cosmological term survives as a matched homogeneous metric ledger. What remains open is the naturalness and matching problem for the measured value, not a failed prediction by a regulator-dependent zero-point sum.

The Standard Model enters at structural theorem level. The SM structural spine yields

$$\Sigma_{SM} = (G_{SM}, R_{SM}, H_{br/fix}, O_Y, Q_{SM}, A_{anom}, V_{SM}, M_{slot}, \Theta_{mix}, N_g) / \simeq_{Adm},$$

with  $G_{SM} = SU(3)_c \times SU(2)_L \times U(1)_Y$ . This is not merely a role schema. It records the recovered gauge-product survivor, chiral representation-response skeleton, Higgs-like bridge/fixation carrier, Yukawa-realizability ledger, charge lattice, anomaly-neutral survivor ledger, vacuum-compatible residual form, mass-response slots, mixing/phase slots, and family-cardinality envelope record. It is also not a full empirical parameter-table closure. Gauge couplings, Yukawa eigenvalues, quark masses, CKM/PMNS coordinates, Higgs potential, vev, scalar mass, self-coupling, weak-scale values, and absolute scale anchors remain downstream parameter-sector targets unless separately certified.

The final posture is therefore neither reductionist overclaim nor loose analogy. The paper does not say that standard physics is unnecessary, false, or merely vocabulary. It says that standard physics becomes physically interpretable through role-bearing interior structure. GR, Einstein dynamics, quantum predictive structure, Yang-Mills local QFT, QFT vacuum-renormalization discipline, and the Standard Model structural tuple are controlled projections or specializations of the realized physical interior. They are not primitive sources for it.

The primitive alternatives are exhausted. A primitive spacetime container lacks non-null standing, carrier mediation, particle identity, gauge discipline, mass response, vacuum baseline, record order, and witness dependence unless those roles are added. A primitive matter inventory lacks physical standing unless its entries instantiate carrier, identity, gauge, deformation, vacuum, and continuation roles. A primitive metric has no boundary standing before comparison is licensed.

Primitive fields, particles, gauge groups, mass scalars, vacuum substances, time parameters, causal relations, information measures, or formal state spaces either import hidden selectors, collapse to skin, change scope, or reproduce the role work already isolated by  $I_D$ .

The open program is equally clear. Numerical constants require scale and calibration ledgers. Full matter-QFT construction requires additional theorem objects beyond the local Yang-Mills import. Quantum gravity requires constraint-native realization certificates rather than primitive projection-to-projection quantization. Thermodynamics, classical limits, scattering theory, cosmological evolution, and detailed parameter sectors remain downstream specializations. These are not defects in the RPI theorem. They are the next layers after the physical interior has been made explicit.

In compressed form:

$$K_D \Rightarrow I_D \Rightarrow \{\text{AER, Gravity, Einstein, Quantum, YM/QFT, ZPE, } \Sigma_{SM}\}.$$

The first arrow is the construction-to-physics role derivation. The second is the standard-physics projection and specialization stack. The realized physical interior is the invariant condition tying the two together: the role-closed architecture through which physical theories become theories of a realized physical regime at all.

## Appendices

### Appendix A - Source-theorem dependency index

This appendix preserves the audit path while keeping the main text continuous. A source may be used only at the claim type declared here. Companion theorem objects are not primitive premises of the realized physical interior, and source disciplines are not promoted beyond their stated scope.

| Source cluster   | Claim type used here             | Function in the manuscript                                                                                            | Boundary                                                            |
|------------------|----------------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| META /<br>KERNEL | mathematical<br>constraint layer | root admissibility,<br>standing,<br>reference,<br>irreversibility,<br>no-repair, no-<br>lower-generator<br>discipline | supplies no<br>physical ontology,<br>metric, fields, or<br>dynamics |

| Source cluster               | Claim type used here                | Function in the manuscript                                                                                                  | Boundary                                                                               |
|------------------------------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| AMetric / ATS                | interface discipline                | no primitive metricity, no representative privilege, anchor/tensor/skin typing                                              | supplies no Einstein equation, QFT, or SM content                                      |
| R1-R12                       | role-source family synthesized here | non-nullity, locality, domains, comparison, transport, fields, particles, gauge, mass, vacuum, time, causality              | component role theorems are source results; this paper integrates them into $I_D$      |
| AER                          | Hamiltonian quotient specialization | standing surface $C$ , quotient map $\pi$ , AER-Obsmin/Sep/Max, $D = \ker(\omega_C)$ , first-class closure, multiplier form | local regular-stratum theorem object; not a premise for the RPI ladder                 |
| GRAV                         | gravity role                        | gravity as forced constraint role of the admissible physical interior                                                       | not a numerical derivation of all gravitational physics                                |
| EIN                          | metric projection                   | SameScopeMetric <sub>1</sub> , normal form $aR_{\mu\nu} + bRg_{\mu\nu} + cg_{\mu\nu}$ , Einstein combination                | no numerical $\kappa$ or $\Lambda$ ; modified/richer scopes are classified, not denied |
| Quantum predictive / records | quantum projection                  | Hilbert/Schrodinger interior, Bell witness typing, measurement-record quotient                                              | no category identity between GR and QM                                                 |

| Source cluster                | Claim type used here                   | Function in the manuscript                                                                                                                          | Boundary                                                                    |
|-------------------------------|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| YM                            | local gauge-QFT specialization         | sharp-local gauge-invariant Yang-Mills local net, OS/Wightman packet, nontriviality, mass gap in declared scope                                     | not all QFT, full SM constructive QFT, matter sectors, or scattering theory |
| ZPE triad                     | renormalization-standing bridge        | $\rho_R$ offset as skin, relational responses as standing, $\Lambda_{eff}$ as bound-ary/matching input                                              | no solution of the naturalness problem                                      |
| SM                            | structural specialization              | $\Sigma_{SM}/\simeq_{Adm}$ , gauge product, representation skeleton, Higgs/bridge, Yukawa-realizability, charge, anomaly, vacuum, mass/mixing slots | no full empirical parameter table                                           |
| Cross-sector / charged lepton | quotient-level numerical extraction    | charged-lepton relation-level mass-shape and one-anchor display status                                                                              | exact pole closure and zero-anchor scale open                               |
| Neutrino standing ratio       | conditional sectoral numerical readout | $\rho_\nu$ window, projective shape, one-anchor atmospheric-gap display                                                                             | zero-anchor absolute mass and PMNS numerical values open                    |

Selected theorem labels for the highest-load uses are listed below.

| Theorem family | Labels tracked in the source material                                                                                                                                                                                                                                                  |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AER            | kernel invocation and denial burden;<br>AER comparison class; redescription<br>distribution; AER-Obsmin; AER-Sep;<br>AER-Max; characteristic capture;<br>exact characteristic identification;<br>first-class closure; multiplier-type<br>dynamics; fixed-domain closure                |
| Yang-Mills/QFT | theorem-packet reduction;<br>compact-simple group scope; flowed<br>continuum state; finite-cap and<br>sharp-local extension;<br>Wilson-to-continuum gap transport;<br>OS/Wightman and field<br>correspondence; faithful-Wilson<br>universality and endpoint                            |
| Standard Model | gauge transition groupoid; connection<br>as standing-preserving transport;<br>curvature/holonomy obstruction;<br>representation and charge rigidity;<br>anomaly ledger; vacuum-compatible<br>residual sector; SR/PSM survivor;<br>Higgs/bridge carrier;<br>Yukawa-realizability ledger |
| Neutrino       | AASCNeutrinoSourceRowStandingAdmission;<br>same-scope admission rows; final<br>numerical readout; projective<br>mass-squared shape; one-anchor<br>reconstruction; atmospheric-gap<br>anchored display                                                                                  |

## Appendix B - Symbol index

| Symbol                            | Meaning                                                             |
|-----------------------------------|---------------------------------------------------------------------|
| $K_D$                             | AASC kernel: admissibility, standing,<br>reference, irreversibility |
| $I_D$                             | realized physical interior                                          |
| $\text{NonVacRec}_D$              | non-vacuous realized persistence                                    |
| $[B_D]_{\equiv B}$                | bounded standing-localization class                                 |
| $\text{Adj}_D, N_D, \text{LCD}_D$ | adjacency, neighborhoods, and local<br>continuation domains         |
| $\text{Cmp}_D$                    | quotient-stable comparison carrier                                  |
| $\text{BoundTrans}_D$             | bounded perturbation transport                                      |

| Symbol                         | Meaning                                                                  |
|--------------------------------|--------------------------------------------------------------------------|
| $\Phi_D$                       | field-carrier role                                                       |
| $\Pi_D$                        | localized particle persistence classes                                   |
| $G_D$                          | gauge role as boundary-fixed<br>presentation equivalence                 |
| $M_D = (\mu_D, \iota_D)$       | mass/inertia deformation-response<br>profile                             |
| $\Omega_D$                     | vacuum baseline role                                                     |
| $T_D$                          | temporal ordering role                                                   |
| $\text{Cau}_D$                 | causal-dependence role                                                   |
| $C$                            | AER standing surface                                                     |
| $\pi : C \rightarrow S_{phys}$ | physical quotient map                                                    |
| $D = \ker(d\pi)$               | redescription distribution                                               |
| $\omega_C$                     | pulled-back symplectic form on the<br>standing surface                   |
| $\Sigma_{SM}$                  | recovered Standard Model structural<br>tuple                             |
| $\rho_\nu$                     | neutrino standing splitting ratio<br>$\Delta m_{21}^2 / \Delta m_{31}^2$ |

## Appendix C - RPI role ladder and deletion matrix

| Role                  | Deletion failure                                                             |
|-----------------------|------------------------------------------------------------------------------|
| $\text{NonVacRec}_D$  | physical realization collapses toward<br>nullity                             |
| $[B_D] \equiv B$      | no bounded locus of standing burden,<br>failure, or restoration              |
| $LCD_D$               | no local continuation domains or<br>adjacency structure                      |
| $\text{Cmp}_D$        | no quotient-stable metric-comparison<br>role                                 |
| $\text{BoundTrans}_D$ | perturbation becomes pathless or<br>global                                   |
| $\Phi_D$              | no distributed persistence carrier                                           |
| $\Pi_D$               | no localized persistence identity                                            |
| $G_D$                 | presentation variation becomes false<br>multiplicity or arbitrary convention |
| $M_D$                 | no deformation-response or inertia<br>profile                                |
| $\Omega_D$            | no non-null no-particle baseline                                             |
| $T_D$                 | no irreversible continuation order                                           |
| $\text{Cau}_D$        | no witness-dependent dependence                                              |

## Appendix D - Standard-physics projection ledger

| Physics object               | Type in this paper                 | Main standing content                                                                            | Principal boundary                               |
|------------------------------|------------------------------------|--------------------------------------------------------------------------------------------------|--------------------------------------------------|
| AER                          | quotient/constraint specialization | coisotropic standing, characteristic redescription, first-class closure, multiplier form         | local regular stratum                            |
| AASC gravity                 | constraint-role projection         | standing surface, quotient observables, admissible redescription, closure, presentation dynamics | not all gravitational dynamics                   |
| Einstein projection          | metric projection                  | minimal first-response Lorentzian normal form                                                    | no numerical constants; richer scopes classified |
| Quantum predictive structure | predictive projection              | Hilbert geometry, Schrodinger dynamics, Bell and record quotient structures                      | no GR/QM category identity                       |
| Yang-Mills/QFT               | local gauge-QFT specialization     | sharp-local gauge-invariant local net and gap packet                                             | pure YM scope, not all QFT                       |
| ZPE discipline               | renormalization-standing bridge    | quotient failure of absolute offsets, relational survivors, $\Lambda_{eff}$ boundary status      | no dark-energy solution                          |
| Standard Model               | structural specialization          | $\Sigma_{SM}/\simeq_{Adm}$                                                                       | no full numerical parameter table                |

## Appendix E - Predictive-status ledger

| Object                            | Reportable status                               | Boundary                                                           |
|-----------------------------------|-------------------------------------------------|--------------------------------------------------------------------|
| RPI role order                    | structural prediction                           | role-theoretic target only                                         |
| Anchor/tensor/skin classification | classificatory prediction                       | depends on fixed-scope typing                                      |
| ZPE absolute offset               | certified obstruction                           | leaves naturalness/matching open                                   |
| Charged-lepton mass-shape         | quotient-level extracted ratio                  | one-anchor MeV display; exact pole closure open                    |
| SM charge-lattice class           | fixed-scope extracted class                     | no $e$ , $\alpha$ , or $g_i$ derivation                            |
| Neutrino splitting ratio          | conditional standing-admission numerical window | source-row admission and one-anchor display; zero-anchor mass open |
| Full SM parameter table           | downstream program                              | requires sector-specific exactness and scale certificates          |

## Appendix F - Route-exhaustion normal forms

A same-scope rival must preserve the fixed target while carrying physically relevant content not already captured by the role architecture. Every such rival falls into one of the following normal forms.

| Normal form                  | Collapse route                                           |
|------------------------------|----------------------------------------------------------|
| anchor motion                | changes the theorem object                               |
| tensor-load import           | declares richer scope or hidden load                     |
| quotient motion              | changes physical identity                                |
| continuation motion          | changes lawful dynamics                                  |
| hidden selector              | introduces a second same-scope classifier                |
| repair / deferred activation | violates irreversibility and no-repair                   |
| representative privilege     | promotes skin to standing content                        |
| metric backflow              | uses a downstream projection to license its own boundary |
| parameter laundering         | promotes calibration or fit into structural proof        |
| skin-only formalism          | changes presentation but no standing-bearing content     |

## Appendix G - Source manifest

The source manifest records the following source clusters: META, KERNEL, R1-R12/R12, AER, GRAV, EIN, YM, ZPE1-ZPE3, SM, cross-sector extrac-

tion, standing neutrino splitting ratio, and standard physics comparators. The manifest gives file names and archive provenance. The main text intentionally uses these sources by theorem family rather than by repeating source inventory throughout the argument.

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